Nonlinear Calculus in Practice 2:

Rigidity, Lebesgue, Solid Deformation, and Multibody Dynamics

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Abstract

This sequel to *Nonlinear Calculus in Practice* develops mathematical connections and worked applications of the Matrix-of-Error framework. Lebesgue integration provides the common foundation linking the earlier probability and quantum-measurement studies to time-dependent budgets, deformation measures, and interacting-body mechanics. A general weighted-budget lemma, abstracted from the earlier enstrophy envelope, supplies equality and quantitative control, with a finite threshold under a persistent decay margin; the periodic flow identifies its rigid configuration. The connection to measured group theory is examined through the work of Margulis, Bader, and Furman, while geometric rigidity supplies a separate, direct connection to solid deformation. An explicit volume-preserving warping map supplies a worked geometric integration. A separate proposed test model couples solid motion, internal strain, and heat through seven stress blocks and companion evolution balances. Worked gravitational and articulated five-body examples retain pair interactions, configuration-dependent inertia, and applied contributions through integration by parts. A controlled comparison finds that replacing resolved bodies by point centers can shift fitted masses by up to 3.205% without improving wider-range held-out gravity predictions. An additional quadrupole correction reduces that mass error to 0.236%; a separate in-range test distinguishes local transfer from extrapolation. Retaining the full volume contribution recovers the known parameters near the observed numerical resolution. A separate leading-quadrupole calculation accounts coherently for radiation from all five bodies. The result is a continuation of the Matrix-of-Error integration process through further mathematical systems, with analytic identities, numerical consistency checks, and prospective physical applications distinguished explicitly.

Keywords: nonlinear integration; Matrix of Error; Lebesgue measure; rigidity; solid deformation; multibody dynamics; parameter identification.

1 Introduction

Nonlinear Calculus in Practice: The Matrix of Error and Cross-System Integration [1] explains how to carry out the integration procedure introduced in the original cross-system study [2]. Its central practical instruction is to retain identifiable physical contributions when passing from a differential balance to its integrated form. A contribution omitted by a nominal description is not thereby a numerical mistake, nor should it automatically be absorbed into a fitted constant. The present

sequel applies that instruction to additional systems and asks what the resulting integrals reveal about constraints, deformation, and interacting bodies.

The progression has four connected stages. First, an envelope is treated as a function that bounds a dynamical quantity, with a precise equality condition and a finite threshold time determined by the budget. Second, volume integration and geometric rigidity describe deformation independently of that fluid example. Third, gravitational interactions are integrated for an arbitrary finite number of point masses and checked on two-, three-, and five-body reference motions. Fourth, the calculation is extended to articulated bodies, resolved mass distributions, and a separate gravitational-radiation example. A controlled comparison then tests whether omitted body structure can be mistaken for different masses.

The interacting-body examples connect two related settings: *gravitational N-body dynamics*, where bodies interact through gravity, and *articulated multibody dynamics*, where mechanical connections also constrain their movement.

Lebesgue measure supplies the shared integration language: probability on a record space, volume and mass within a body, boundary traction on a surface, or a dynamical contribution over a time interval. The earlier binary-measurement studies [3, 4, 5] provide the quantum-to-calculus starting point. Here that connection is carried into further mathematical applications.

1.1 Method and contribution

The contribution carried forward from [2, 1] is the independently developed Matrix-of-Error framework. It organizes integration by parts and identifiable physical contributions in a shared process demonstrated across nonlinear systems and dimensions. The process specifies a nominal balance, calculates the additional physical terms, retains their signed integral contributions, and checks both closure and the consequences of omitting a contribution. Nonlinear interactions remain in those terms even when unknown coefficients enter the resulting matrix equations linearly.

For a row indexed by a test window and an equation component, write the signed contributions as

$$\mathcal{C}_{k\ell} = \int_{I_k} c_\ell(t)\phi_k(t)\,\mathrm{d}t, \qquad r_k = \sum_\ell \mathcal{C}_{k\ell}, \tag{1.1}$$

with derivative terms transferred to the test function where appropriate. The *Matrix of Error* records the identified correction columns relative to the stated nominal description. The full contribution array also contains the nominal terms. Its row sum is zero for an exact complete balance; a nonzero numerical closure residual is a separate diagnostic. Physical additions may cancel in their total without individually vanishing.

The computational comparison holds the integration and fitting process fixed while changing the representation of mutual gravity. This controlled difference makes it possible to trace a parameter shift to a specific omitted contribution. The original point-versus-volume comparison is supplemented by a second-moment correction and an independent in-range test in Section 8.7.

Related methods. Integration by parts, weak test functions, and parameter estimation from integrated equations also appear in Messenger and Bortz's WSINDy work [6, 7], the SPIDER framework of Gurevich and collaborators [8], and WENDy [9]. These provide similar, though

independent methodological precedents. The present study applies the Matrix-of-Error organization and process to the additional systems as application examples below.

The existing laws and reference theorems are not challenged. This paper presents additional worked examples.

2 Lebesgue Measure as the Shared Foundation

2.1 From binary records to integration

The earlier probability series [3, 4, 5] connects binary records, probability measures, and definite integrals. The shared measure identity is the starting point for the applications here. If the coordinates b_k are independent fair bits under a product probability $\mathbb{P}$, the binary map

$$\pi(b) = \sum_{k=1}^{\infty} b_k 2^{-k}, \qquad \pi_*\mathbb{P} = \mathcal{L}^1|_{[0,1]}, \qquad \mathbb{E}_{\mathbb{P}}[f(\pi(b))] = \int_0^1 f(x)\,\mathrm{d}x \tag{2.1}$$

holds for integrable f. A length-n cylinder has probability 2^{-n} and corresponds to a dyadic interval of that length. The two expansions at dyadic endpoints affect a null set. The identity relates the measure on complete records to the measure on their images; finite records specify the corresponding dyadic intervals.

In the repeated quantum-nondemolition measurement setting examined in [5], the asymptotic outcome weights and the assumptions on the measurement model remain essential. Independent resets of a complete, symmetrically prepared two-outcome experiment can supply fair endpoint records. Successive observations within one such experiment are not thereby independent fair bits. This distinction keeps the quantum origin of the discussion explicit while identifying the measure-theoretic operation used subsequently.

2.2 Physical totals and normalized expectations

For a material body $\mathcal{B}$ with finite volume and reference density ρ_0, define

$$|\mathcal{B}| = \int_{\mathcal{B}} \mathrm{d}X, \quad m = \int_{\mathcal{B}} \rho_0\,\mathrm{d}X, \quad \mathrm{d}\mu_V = \frac{\mathrm{d}X}{|\mathcal{B}|}, \quad \mathrm{d}\mu_m = \frac{\rho_0\,\mathrm{d}X}{m}. \tag{2.2}$$

Then $\int_{\mathcal{B}} f\,\mathrm{d}X = |\mathcal{B}|\mathbb{E}_{\mu_V} f$ and $\int_{\mathcal{B}} f\rho_0\,\mathrm{d}X = m\mathbb{E}_{\mu_m} f$. Normalization permits expectation notation; the prefactors restore physical units. A body's mass measure need not arise from random motion or from a quantum preparation.

The measure also changes with the object being integrated. Point masses use a discrete measure $\sum_i m_i\delta_{r_i}$, not a three-dimensional volume density. A resolved body uses $\rho\,\mathrm{d}V$. Traction is integrated against surface area on its boundary, which has zero volume but can transmit nonzero force. Time-dependent balances use $\mathrm{d}t$, or product measures such as $\mathrm{d}X\,\mathrm{d}t$. Fubini's theorem applies under absolute integrability; Tonelli's theorem applies to nonnegative integrands. These are conditions on integration, not a physical separation of the space and time involved in the motion.

One elementary consequence will be particularly useful: if $f \geq 0$ and $\int f\,\mathrm{d}\mu = 0$, then $f = 0$ almost everywhere. In the envelope calculation this turns zero loss of a weighted budget into zero

dissipation. In deformation it turns zero integrated distance from rotations into a rigid configuration, after the geometric compatibility hypotheses are supplied.

2.3 Rigidity and quantitative stability

The budget and deformation sections share a two-level structure: an equality case identifies a rigid configuration, and a quantitative estimate controls departure from that case. In the periodic-flow instance of Section 3, zero weighted-budget loss forces a spatially constant flow, while small loss controls the weighted integral of enstrophy. In Section 4, zero integrated distance of a compatible deformation gradient from $SO(n)$ forces one rigid motion, $y = RX + c$; the Friesecke–James–Müller estimate (4.2) controls departure from one rotation when that distance is small. Lebesgue integration measures accumulated departure over time or volume. Nonnegativity makes zero accumulated departure an almost-everywhere statement; the balance law or geometric compatibility then identifies the rigid configuration. Quantitative inequalities turn small departures into controlled proximity in the stated integral norms.

This equality-and-stability parallel connects the applications without identifying their equations. The geometric estimate is static; a time-decay rate additionally requires an evolution law and a coercive dissipation bound, and the budget's finite threshold requires a persistent margin over its majorant. Viscous dissipation does not automatically control a solid's stored strain energy, since a stationary deformation can retain energy without viscous loss. Each proposed energy and loading regime must satisfy the relevant hypotheses.

Superrigidity concerns representations or cocycles of specified groups and actions. It remains distinct from the equality and geometric rigidity just described: a shared integration theory does not supply its group-theoretic hypotheses. Section 3 examines the measured structures without assuming a transfer of those theorems.

3 Envelopes, Equality Cases, and Measured Dynamics

3.1 An envelope obtained from a budget

An *upper envelope* for a nonnegative quantity $Y(t)$ on an interval I is a function $U(t)$ satisfying $Y(t) \le U(t)$ there. It need not be an exact trajectory, an asymptote, or the smallest possible bound. A useful envelope carries assumptions that can be checked from the balance producing it. The following argument abstracts the budget structure of the earlier Navier–Stokes continuation envelope [2, 10]. Its weighted loss identity and conditional threshold estimates apply to any system satisfying the stated balance, majorant, and coercivity assumptions. The periodic flow below identifies the equality configuration in one concrete instance.

On $I = [0, T)$, let $Y \ge 0$ be locally absolutely continuous, and let B and $C \ge 0$ be locally integrable. Assume the exact balance

$$\frac{1}{2}Y' = B - C \qquad \text{almost everywhere.} \tag{3.1}$$

Fix $0 < \delta < 1$ and let $g \ge 0$ be measurable and locally integrable, with a valid majorant

$$B \le (1 - \delta)C + gY. \tag{3.2}$$

For the quantitative control of Y, assume also $C \geq \lambda_c Y$ almost everywhere for a fixed $\lambda_c > 0$. The ingredients are a nonnegative state, nonnegative dissipation, an exact balance, a valid majorant, and this coercivity bound. Nonnegativity of g alone does not make it a valid majorant. Put

$$A_g(t) = \int_0^t g(s)\,\mathrm{d}s, \qquad H_g(t) = e^{-2A_g(t)}, \qquad U_g(t) = Y(0)e^{2A_g(t)}. \tag{3.3}$$

Integrating the weighted differential inequality gives

$$H_g(t)Y(t) + 2\delta \int_0^t H_g(s)C(s)\,\mathrm{d}s \leq Y(0), \qquad Y(t) \leq U_g(t). \tag{3.4}$$

The simple envelope U_g is nondecreasing when $g \geq 0$, even if the actual Y decreases. A descending trajectory and an increasing upper bound are compatible. The weighted inequality itself needs no coercivity assumption; $C \geq \lambda_c Y$ supplies the quantitative control below.

3.2 Equality rigidity and quantitative control

Define the estimate slack and the weighted state by

$$q_g = (1-\delta)C + gY - B \geq 0, \qquad Z = H_g Y. \tag{3.5}$$

The slack is the difference between an estimate and the quantity estimated. It is not an extra physical force or a correction column of the Matrix of Error.

Lemma 3.1 (General weighted budget)**.** *Under the assumptions of Section 3.1, Z is locally absolutely continuous and nonincreasing. For $0 \leq a < b < T$, its loss is*

$$Z' = -2H_g(\delta C + q_g), \qquad Z(a) - Z(b) = 2\int_a^b H_g(\delta C + q_g)\,\mathrm{d}t. \tag{3.6}$$

Equality $Z(a) = Z(b)$ holds if and only if $C = q_g = 0$ almost everywhere on $[a, b]$. With coercivity it also forces $Y = 0$ throughout that interval. Quantitatively,

$$\int_a^b H_g Y\,\mathrm{d}t \leq \frac{Z(a) - Z(b)}{2\delta\lambda_c}, \tag{3.7}$$

$$Z(b) \leq e^{-2\delta\lambda_c(b-a)} Z(a),$$

$$Y(b) \leq Y(a)\exp\left(2\int_a^b g\,\mathrm{d}t - 2\delta\lambda_c(b-a)\right). \tag{3.8}$$

Proof. Since $H_g' = -2gH_g$, the product rule and (3.1) give $Z' = 2H_g(B - C - gY) = -2H_g(\delta C + q_g)$. Integration gives (3.6). On any compact interval $H_g > 0$, so the nonnegative-integral property of Lebesgue integration gives the equality characterization. Coercivity and continuity then give $Y = 0$ on the interval. Using $C \geq \lambda_c Y$ in the loss identity proves (3.7); the differential inequality $Z' \leq -2\delta\lambda_c Z$ proves (3.8). □

The first estimate is quantitative near-equality control, integrated over time; it is not a pointwise bound on every intermediate state. The last estimate separates control of the weighted Z from decay of the unweighted Y. These consequences belong to the budget structure and do not depend on the exponent or component choice used to obtain g. Identifying a physical configuration from zero dissipation requires the particular system's structure.

Lyapunov–LaSalle context. The use of a decreasing quantity and its zero-loss states belongs to the Lyapunov–LaSalle tradition. For a smooth autonomous finite-dimensional system on a compact positively invariant set, LaSalle's principle relates a continuously differentiable V with $\dot{V} \leq 0$ to asymptotic approach to the largest invariant subset of $\{\dot{V} = 0\}$ [39, Theorem 1]. The present Z may depend on accumulated history through A_g, so that principle is context rather than an invoked theorem for this budget. Here the exact loss identity proves the finite-interval equality statement directly; coercivity and a persistent decay margin supply the explicit threshold in Section 3.3.

Periodic-flow instance. Let u be a smooth, unforced, incompressible periodic flow on a connected flat torus Ω, with viscosity $\nu > 0$, vorticity $\omega = \nabla \times u$, and strain $S = (\nabla u + \nabla u^T)/2$. Its enstrophy balance has

$$Y = \|\omega\|_2^2, \qquad B = \int_\Omega \omega^T S \omega \, \mathrm{d}x, \qquad C = \nu \, \|\nabla \omega\|_2^2 .$$

A periodic curl has zero spatial mean. If $\lambda_1 > 0$ is the first positive eigenvalue of $-\Delta$, then $C \geq \nu \lambda_1 Y$, so the lemma applies with $\lambda_c = \nu \lambda_1$.

Corollary 3.2 (Equality rigidity of the periodic budget)**.** *For this flow and any valid nonnegative majorant g in* (3.2)*, equality $Z(a) = Z(b)$ forces the flow to be spatially constant on $[a, b]$. Conversely, a spatially constant flow saturates this weighted-state comparison.*

Proof. The lemma gives $Y = 0$, hence $\omega = 0$. A divergence-free and curl-free periodic vector field is spatially constant. Smoothness extends the almost-everywhere conclusion across the interval, and conservation of the mean for the unforced equation makes that constant independent of time. Conversely, a spatially constant flow has $Y = C = B = 0$, and hence $Z = 0$. □

The earlier integration connects this balance to the two-component continuation criterion of Chae and Choe [11], through a blockwise estimate with coefficient proportional to $\nu^{-3} \|P_\chi \omega\|_2^4$, where P_χ projects onto the plane perpendicular to a fixed unit direction χ. For that criterion, control is required through each finite horizon under examination; it does not impose a finite value of the total integral on $[0, \infty)$.

3.3 A finite threshold time determined by the budget

Suppose that, from t_0 onward on the interval of validity, $g(t) \leq \delta \lambda_c - \alpha$ almost everywhere for some $\alpha > 0$. Then $Y(t) \leq Y(t_0) e^{-2\alpha(t - t_0)}$. For a specified threshold $\varepsilon > 0$ and $Y(t_0) > 0$, define

$$T_\varepsilon = t_0 + \frac{1}{2\alpha} \max \left\{ 0, \log \frac{Y(t_0)}{\varepsilon} \right\}. \tag{3.9}$$

If $Y(t_0) = 0$, take $T_\varepsilon = t_0$. Subject to the persistent bound, the function is at or below ε for all later times in that interval. This provides a mathematically controlled finite scope for a specified degree of completion, without evaluating an asymptote indefinitely. It is a guaranteed threshold time, not necessarily the first crossing, and it does not turn asymptotic zero into an exact finite-time zero.

The positive margin α is essential to this estimate; boundedness of g alone is insufficient. For the flow instance, this is a sufficient decay condition, not an equivalence with smooth regularity. The

general threshold uses no fluid-specific equation once the budget hypotheses and persistent margin hold.

For the periodic-flow coefficient inherited from the earlier calculation, set $\lambda_c = \nu\lambda_1$ and write $g = K\,\|P_\chi\omega\|_2^4 \le KY^2$ with $K > 0$, and set $c = \delta\nu\lambda_1$. If

$$0 < Y(t_0) < \sqrt{c/K}, \qquad \alpha = c - KY(t_0)^2 > 0, \tag{3.10}$$

a first-exit argument gives $Y(t) \le Y(t_0)$ while the solution and budget remain valid. The required bound on g then persists, and (3.9) follows. This is an explicit sufficient condition supplied by the state; the calculation does not assume that every initial state enters this regime.

This recovers the classical small-enstrophy decay and regularity mechanism in the retained budget form [40, Section 6.4]; see also [35]. In the original normalization $K = K_0\nu^{-3}$, its threshold is $Y(t_0) < \nu^2\sqrt{\delta\lambda_1/K_0}$. The additional purpose of the calculation is to make the associated threshold time explicit.

3.4 The relationship to Margulis, Bader, and Furman

The published literature supplies precise structures with which to compare this envelope. Margulis's normal-subgroup theorem [12] and Bader–Shalom's factor and normal-subgroup results [13] constrain subgroups and actions under specified higher-rank or product-group assumptions. Bader and Furman's extension of superrigidity [14] concerns extension of suitable lattice representations to an ambient group. Furman's survey [15] explains the measure-equivalence and orbit-equivalence setting. In this context a lattice is a discrete subgroup with a finite invariant quotient measure.

Lebesgue differentiation also enters the analysis of these measured structures. For example, Bader, Furman, and Sauer use it in their analysis of boundary maps in integrable measure equivalence [16, Lemma 3.8]. Here the comparison focuses on measured evolution and composition; the finite-time conclusions follow from the budget's loss and decay estimates.

A concrete connection available here is *composition along a measured evolution*. Define

$$\mathfrak{m}(t,s) = \exp\left(2\int_s^t g(\tau)\,\mathrm{d}\tau\right), \qquad \mathfrak{m}(t,r)\mathfrak{m}(r,s) = \mathfrak{m}(t,s). \tag{3.11}$$

If g is evaluated along an autonomous evolution Φ_t, this is the multiplicative cocycle rule $\mathfrak{m}(t+s,v) = \mathfrak{m}(t,\Phi_s v)\mathfrak{m}(s,v)$. For a smooth incompressible flow, the particle map $X(t,s,a)$ also satisfies

$$\det D_a X(t,s,a) = 1, \qquad D_a X(t,r,X(r,s,a))D_a X(r,s,a) = D_a X(t,s,a). \tag{3.12}$$

Thus volume measure is preserved, while the derivative records stretching. Volume preservation does not require preservation of lengths or enstrophy.

The common structures are measured evolution and cocycle composition. Applying a group-theoretic superrigidity theorem would additionally require the specified group action and representation hypotheses. The conclusions established here are the general budget lemma, its equality-rigidity corollary, and the transient estimates above.

3.5 Why the retained stationary route is trivial

There is also a useful obstruction. Suppose a globally defined measurable evolution on finite-energy periodic states satisfies the unforced energy inequality and preserves the spatial mean. Let $e(v) = \frac{1}{2}\|v - \bar{v}\|_2^2$. Poincare's inequality gives $e(\Phi_t v) \le e^{-2\nu\lambda_1 t} e(v)$. If a probability μ on this state space is invariant and $e < \infty$ almost surely, then for $r > 0$,

$$\mu\{e > r\} = \mu\{e \circ \Phi_t > r\} \le \mu\{e > e^{2\nu\lambda_1 t} r\} \longrightarrow 0. \tag{3.13}$$

No finite ensemble mean of e is required for this tail argument. The invariant probability is supported on spatially constant flows. Their particle maps are translations, with identity spatial derivative. Hence this retained unforced setting does not yield a nontrivial stationary stretching cocycle. The conditional statement assumes the evolution just specified; it does not assert global smooth existence for arbitrary data. For the current application, the useful information lies in transient loss, near-equality, and function-dependent scope.

4 Solid Deformation and Volume Integration

Solid deformation provides a separate application of the integration framework. Its purpose is to propose a physically richer coupled problem and investigate whether organizing its contributions through Nonlinear Calculus makes it more tractable. A solid requires a configuration, a stress response, loading, and any material history needed to describe recovery or permanent change. The completed calculation below establishes the volume measures and a worked warping geometry. A separate proposed formulation then retains motion, internal deformation, and thermal feedback explicitly; its constitutive equations and integrated structure are specified, while a material-specific prediction remains to be tested.

4.1 Configuration, mass, and rigidity

Let $\mathcal{B} \subset \mathbb{R}^n$, $n \ge 2$, be a bounded, connected Lipschitz reference body, with deformation $y(X,t) \in \mathbb{R}^n$, velocity $v = \partial_t y$, gradient $F = \nabla_X y \in \mathbb{R}^{n\times n}$, and volume ratio $J = \det F$. The general formulation leaves n unspecified; the explicit warp below specializes to $n = 3$. For an injective orientation-preserving configuration, the change of variables and conservation of mass read

$$\int_{y(\mathcal{B},t)} f(x)\,\mathrm{d}x = \int_{\mathcal{B}} f(y(X,t))J\,\mathrm{d}X, \qquad \rho(y(X,t),t)J = \rho_0(X). \tag{4.1}$$

The kinetic energy is $\frac{1}{2}\int_{\mathcal{B}} \rho_0|v|^2\,\mathrm{d}X$. A stored-energy density $W(X,F,\vartheta,z)$ per reference volume gives $\int_{\mathcal{B}} W\,\mathrm{d}X$, where ϑ denotes temperature and z any internal variables. If a density per unit mass is used instead, the factor ρ_0 must be included. The continuum balance and constitutive setting are standard [17].

Friesecke, James, and Müller prove that for every $y \in W^{1,2}(\mathcal{B};\mathbb{R}^n)$ there is a single $R \in SO(n)$ such that [18, Theorem 3.1]

$$\|\nabla y - R\|_{L^2(\mathcal{B})} \le C_{\mathcal{B}} \|\mathrm{dist}(\nabla y, SO(n))\|_{L^2(\mathcal{B})}. \tag{4.2}$$

The right side measures pointwise departure from rotations; the left controls departure from one common rotation across the body. If the right side vanishes, $y = RX + c$ almost everywhere. With a coercive stored energy satisfying $W \geq \kappa \operatorname{dist}^2(F, SO(n))$, this also yields

$$\int_{\mathcal{B}} |F - R|^2 \, \mathrm{d}X \leq \frac{C_{\mathcal{B}}^2}{\kappa} \int_{\mathcal{B}} W \, \mathrm{d}X. \tag{4.3}$$

This geometric rigidity is a direct relation between an integrated deformation measure and a rigid configuration. Its constant depends on the domain, and it does not supply an evolution law or a spectral gap.

The linear rigidity comparison. For a displacement $\xi \in H^1(\mathcal{B}; \mathbb{R}^n)$, Korn's inequality gives a constant skew-symmetric matrix A such that

$$\|\nabla \xi - A\|_{L^2(\mathcal{B})} \leq C_{\mathcal{B}}^{\mathrm{K}} \left\| \tfrac{1}{2}(\nabla \xi + \nabla \xi^T) \right\|_{L^2(\mathcal{B})}.$$

Thus zero infinitesimal strain forces $\xi = AX + b$, an infinitesimal rigid motion, and small integrated strain controls departure from one such motion. This is the linear counterpart of (4.2), whose rigid set consists of finite rotations. Both use Lebesgue norms across the body, not isolated point measurements. Conti, Dolzmann, and Müller state this correspondence and extend the estimates to mixed $L^p + L^q$ growth [38]. This comparison places linear and nonlinear rigidity in the same integral setting. The deformation and constitutive laws below retain their nonlinear form.

Frame indifference requires invariance under a superposed rigid rotation, $W(RF) = W(F)$. It is distinct from isotropy, which concerns rotations of the material reference. If heating or growth changes a natural distortion $A_0(X, t)$, the elastic distortion may be FA_0^{-1}. This field need not itself be a gradient, so (4.2) cannot be transferred to it without checking compatibility. Temperature-dependent material geometry [19] and internal plastic metrics [20] are relevant established routes for specifying such effects.

4.2 The weak momentum balance

Let P be the first Piola stress, b the body acceleration, and t_N the prescribed traction on the Neumann portion Γ_N of the reference boundary. The equations are

$$\rho_0 \partial_t v = \operatorname{Div}_X P + \rho_0 b, \qquad PN = t_N \quad \text{on } \Gamma_N. \tag{4.4}$$

Take a vector test function φ that vanishes at the temporal endpoints and on the Dirichlet boundary. Integration by parts in time and in reference volume gives

$$\begin{aligned} -\int_I \int_{\mathcal{B}} \rho_0 v \cdot \partial_t \varphi \, \mathrm{d}X \, \mathrm{d}t + \int_I \int_{\mathcal{B}} P : \nabla_X \varphi \, \mathrm{d}X \, \mathrm{d}t - \int_I \int_{\mathcal{B}} \rho_0 b \cdot \varphi \, \mathrm{d}X \, \mathrm{d}t \\ - \int_I \int_{\Gamma_N} t_N \cdot \varphi \, \mathrm{d}S \, \mathrm{d}t = 0. \end{aligned} \tag{4.5}$$

Without temporal endpoint cancellation, the additional term is $[\int_{\mathcal{B}} \rho_0 v \cdot \varphi \, \mathrm{d}X]_{\partial I}$. Surface measure $\mathrm{d}S$ retains loading that cannot be represented by a volume integral over the boundary. Choosing interior test functions removes the boundary term from that particular identity; it does not remove the physical load from the motion.

For a selected stress decomposition $P = P_0 + \sum_j c_j P_j(F, \partial_t F, \vartheta, z)$, allowing rate dependence where required, the stress columns are

$$\Theta_{kj} = \int_I \int_{\mathcal{B}} P_j : \nabla_X \varphi_k \, \mathrm{d}X \, \mathrm{d}t, \qquad d_k + \sum_j c_j \Theta_{kj} = 0, \tag{4.6}$$

where d_k contains the other signed terms of (4.5). Nonlinear dependence on the state can remain inside a column even when the unknown coefficients enter linearly. This is the same physical-contribution organization used in the methods bridge [1].

There is an important identifiability restriction. For a smooth deformation, the Piola identity gives $\mathrm{Div}(\mathrm{cof}\,\nabla y) = 0$. A constant-coefficient stress term proportional to $\mathrm{cof}\,\nabla y$ therefore has zero column against interior tests after integration by parts. More interior windows cannot identify that coefficient. Boundary traction information or other suitable observations are needed. This is a structural limitation of the observations, not a defect corrected by adding numerical precision.

4.3 An explicitly integrated warping map

As a retained $n = 3$ verification case, on $\mathcal{B} = (-L/2, L/2)^3$, consider

$$y(X, t) = \big(X_1 + a(t) X_2^2,\ X_2,\ X_3\big), \qquad F = I + 2aX_2 e_1 \otimes e_2, \qquad J = 1. \tag{4.7}$$

This is a position-dependent simple shear. The amplitude a has units of inverse length. The map is globally invertible, with inverse obtained by subtracting aX_2^2 from the first component, and preserves volume. It nevertheless changes lengths and angles.

Set $q = 2aX_2$. The Green strain and its squared Frobenius norm are

$$E_G = \frac{F^T F - I}{2} = \begin{pmatrix} 0 & q/2 & 0 \\ q/2 & q^2/2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad |E_G|^2 = 2a^2 X_2^2 + 4a^4 X_2^4. \tag{4.8}$$

The volume integral is therefore explicit:

$$\mathcal{D}(t) = \int_{\mathcal{B}} |E_G|^2 \, \mathrm{d}X = \frac{a(t)^2 L^5}{6} + \frac{a(t)^4 L^7}{20}. \tag{4.9}$$

The fourth-order contribution remains part of the calculation. $\mathcal{D}$ has units of volume; it becomes an energy only after a material law with the appropriate units is supplied.

For this shear gradient, the pointwise distance to rotations can also be evaluated:

$$\mathrm{dist}^2(F, SO(3)) = q^2 + 4 - 2\sqrt{q^2 + 4} = \frac{q^2 \sqrt{q^2 + 4}}{\sqrt{q^2 + 4} + 2}. \tag{4.10}$$

The last form avoids subtracting nearly equal terms when q is small. Its bounds are $q^2/2 \le \mathrm{dist}^2(F, SO(3)) \le q^2$. By symmetry $\int_{\mathcal{B}} F \, \mathrm{d}X = |\mathcal{B}| I$, so the best constant rotation is I and

$$\min_{R \in SO(3)} \int_{\mathcal{B}} |F - R|^2 \, \mathrm{d}X = \frac{a^2 L^5}{3}, \qquad \mathcal{D}_{SO} \le \frac{a^2 L^5}{3} \le 2\mathcal{D}_{SO}, \tag{4.11}$$

where $\mathcal{D}_{SO} = \int_{\mathcal{B}} \mathrm{dist}^2(F, SO(3)) \, \mathrm{d}X$. This gives a checked family-level comparison between local rotational defect and global rigidity, separately from the fluid envelope.

What happens physically. Different layers of the body shift by different amounts in the first direction. Equal volume does not mean equal shape. The quadratic and quartic terms in (4.9) quantify that distortion. Returning $a(t)$ to zero restores this reference shape; retaining a nonzero final amplitude retains a warped configuration. The formula measures either history, but the cause of that history must come from the stress, thermal input, constraints, and material response.

4.4 Time-linked deformation, rotation, and material resistance

The one-amplitude warp is a worked reduction, not a restriction on the number of coupled deformation components or material variables. The full configuration $y(X,t)$, temperature $\vartheta(X,t)$, and internal variables $z(X,t)$ belong to the same evolving body and time interval. Their equations must retain the components, constraints, and couplings required by the material and loading. A reduced description may use several vector-valued modes,

$$y(X,t) = X + \sum_{\alpha=1}^{m} a_\alpha(t)\psi_\alpha(X), \tag{4.12}$$

with boundary-compatible $\psi_\alpha : \mathcal{B} \to \mathbb{R}^n$ and amplitudes that preserve admissibility of the configuration. Neither n nor the required number m is fixed by the integration process. Each amplitude evolves with time, and its coupling to the others remains in the stress and weak balance; additional material variables require their own compatible evolution equations.

To distinguish deformation from local rotation, express the material velocity in the current configuration as $u(x,t) = v(y^{-1}(x,t),t)$. Standard continuum kinematics [21] gives

$$\mathsf{L} = \nabla_x u = (\partial_t F)F^{-1}, \qquad \mathsf{D} = \tfrac{1}{2}(\mathsf{L} + \mathsf{L}^T), \qquad \Omega = \tfrac{1}{2}(\mathsf{L} - \mathsf{L}^T), \tag{4.13}$$

where the material quantities on the right are evaluated at $X = y^{-1}(x,t)$. The $n \times n$ tensor D measures stretching and shear rates, while Ω measures local spin. For the $n = 3$ example, in a fixed inertial frame, vorticity $\omega = \nabla_x \times u$ and enstrophy quantify rotational motion, not stored distortion. Direct calculation for (4.7) gives

$$\begin{aligned} u(x,t) &= \dot{a}(t)x_2^2 e_1, \qquad \omega(x,t) = -2\dot{a}(t)x_2 e_3, \\ \mathcal{Z}(t) &= \frac{1}{2}\int_{y(\mathcal{B},t)} |\omega|^2 \,\mathrm{d}x = \frac{\dot{a}(t)^2 L^5}{6}. \end{aligned} \tag{4.14}$$

Here $(\omega \cdot \nabla_x)u = 0$: the map deforms without vortex stretching. If $\dot{a} = 0$ while $a \neq 0$, its enstrophy vanishes but its deformation measure (4.9) remains positive. Thus motion and retained shape require distinct, time-linked measures. For general n, the skew tensor Ω remains available without using a three-component curl; the rotational measure $\int_{y(\mathcal{B},t)} |\Omega|^2 \,\mathrm{d}x$ agrees with $\mathcal{Z}$ when $n = 3$.

Material resistance also has distinct contributions. Elastic response stores recoverable energy; viscous response dissipates energy during deformation; an internal-variable law can describe irreversible change. Nonlinear viscoelasticity supplies established constitutive and evolution equations for such coupled responses [22]. At finite deformation, elastic and dissipative stresses must use compatible stress measures and respect frame indifference [23].

For example, restrict attention to an isothermal material without evolving internal variables, with $P_{\mathrm{el}} = \partial W/\partial F$ and a dissipative stress P_{dis} satisfying $P_{\mathrm{dis}} : \partial_t F \geq 0$. Multiplying the momentum

balance by v and integrating gives

$$\mathcal{E}(t) = \frac{1}{2}\int_{\mathcal{B}} \rho_0 |v|^2 \,\mathrm{d}X + \int_{\mathcal{B}} W(F)\,\mathrm{d}X,$$
$$\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}t} + \int_{\mathcal{B}} P_{\mathrm{dis}} : \partial_t F \,\mathrm{d}X = \mathcal{P}_{\mathrm{ext}}(t), \tag{4.15}$$

where $\mathcal{P}_{\mathrm{ext}}$ includes body-force power and boundary work, including any moving supports. This budget expresses the competition between loading, stored energy, and dissipation. Heating and evolving internal variables require the corresponding thermal and material balances. The elastic and dissipative stresses supply separate contributions to (4.6); neither is replaced by an enstrophy measurement.

What happens physically. Loading drives different parts of the body to move relative to one another. The material can oppose that motion, store energy, dissipate energy, or change its internal state. Warping and subsequent recovery or permanent change depend on that coupled evolution. All required displacement components and material variables remain linked to time, even when a worked example reduces their behavior to a single amplitude.

4.5 Proposed coupled solid-deformation formulation

The proposed target has more retained physical mechanisms than the earlier unforced fluid example: elastic storage, rate-dependent resistance, internal deformation, heat transport, and thermomechanical feedback all contribute to the motion. The hypothesis is that explicit integrated contributions can make this richer problem easier to calculate and interpret. The following candidate makes that hypothesis testable without prescribing a warping amplitude. Established finite-strain thermoviscoelasticity supplies the energy, frame-indifference, and heat-transfer requirements [24]; the particular constitutive family specified here supplies a concrete illustrative test model.

Coupled fields and constitutive choice. On the reference body $\mathcal{B} \subset \mathbb{R}^n$, retain $y(X,t) \in \mathbb{R}^n$, a symmetric internal deformation tensor $Z(X,t) \in \mathbb{R}^{n\times n}$, and temperature $\vartheta(X,t) > 0$. The proposed system is indexed by arbitrary finite $n \geq 2$, including $n \geq 4$; $n = 3$ is only the retained verification example. There are $n + n(n+1)/2 + 1$ scalar field components before introducing velocities as separate first-order variables, giving 15 when $n = 4$. All evolve on the same time interval. Field components and physical-contribution matrices count different aspects of the formulation. With $I = I_n$, let

$$E = \tfrac{1}{2}(F^T F - I), \qquad A = E - Z, \qquad \operatorname{dev} A = A - \frac{\operatorname{tr}(A)}{n} I. \tag{4.16}$$

The configuration retains every entry of F and its cross-component interactions; it is not restricted to a single bending or warping amplitude. The required n, number of deformation modes in (4.12), and number of internal material variables must be selected for the body and loading. Additional deformation modes or material fields can be retained with their own coupled evolution, work, and dissipation contributions. The following y, Z, ϑ system is an explicit starting family.

Here Z records internal strain relative to the reference body, rather than an independent displacement. Choose a free energy per reference volume

$$\Psi(F,Z,\vartheta) = \mu_s|\operatorname{dev} A|^2 + \frac{\kappa_s}{2}(\operatorname{tr} A)^2 + \frac{H_s}{2}|Z|^2 - \beta(\vartheta-\vartheta_0)\operatorname{tr} A \\ + c_0\left[(\vartheta-\vartheta_0) - \vartheta\log(\vartheta/\vartheta_0)\right]. \tag{4.17}$$

Take $\mu_s, \kappa_s, c_0, \vartheta_0 > 0$, $H_s \geq 0$, and a stated thermal-coupling coefficient β. The moduli μ_s, κ_s, H_s have units of stress, while β and c_0 have units of energy per reference volume per temperature. The selected reference-strain energy must be assessed over the intended strain and temperature range; it does not enforce $J > 0$ by itself. Since E is unchanged by a superposed rigid rotation, this choice is frame indifferent. Its thermal capacity is $-\vartheta\Psi_{\vartheta\vartheta} = c_0$.

The quadratic Green-strain elastic part is of St. Venant–Kirchhoff type: frame indifference alone does not give global convexity or a barrier against $J \to 0$, which can occur at finite stored energy. The existence theorem in [24] uses second-gradient regularization, determinant control, and a quasistatic setting. It does not establish existence for the present inertial internal-variable candidate. The onset calculation below consequently states its smooth-solution assumption explicitly.

Define elastic and viscous second-Piola contributions, an internal driving tensor, and dissipation by

$$\begin{aligned} S_e &= 2\mu_s \operatorname{dev} A + \kappa_s \operatorname{tr}(A) I - \beta(\vartheta-\vartheta_0)I, \\ S_v &= 2\eta_s \operatorname{dev}\dot{E} + \zeta_s \operatorname{tr}(\dot{E})I, \qquad P = F(S_e + S_v), \\ R &= S_e - H_s Z, \qquad \dot{Z} = \gamma R, \\ d_v &= 2\eta_s|\operatorname{dev}\dot{E}|^2 + \zeta_s(\operatorname{tr}\dot{E})^2, \qquad d_Z = \gamma|R|^2. \end{aligned} \tag{4.18}$$

The dot denotes a derivative at fixed X, with $\dot{E} = \operatorname{sym}(F^T\nabla_X v)$. The viscosities $\eta_s, \zeta_s \geq 0$ have units of stress times time, and mobility $\gamma > 0$ has units of inverse stress per time. This internal law is a dissipative relaxation choice, $\dot{Z} = -\gamma\Psi_Z$. A yield law or additional internal tensors can supplement or replace it when required by the material; each extension must retain its own work and dissipation terms.

In its isothermal small-strain limit, the internal relaxation branch has standard-linear-solid (Zener) form when $H_s > 0$, with a Maxwell-type limit at $H_s = 0$; S_v adds a parallel strain-rate viscous response. Thus the complete candidate is not merely a three-element Zener model. Here $E - Z$ is a declared reference-strain constitutive choice. Multiplicative elastic–viscous distortion models [22] provide an established alternative at finite deformation; the present candidate uses the declared reference-strain description, with equivalence to a multiplicative model requiring a separate analysis.

For constant current-configuration conductivity $k > 0$, set $K = kJF^{-1}F^{-T}$ and $Q = -K\nabla_X\vartheta$. With r denoting heat supply per reference volume, the coupled equations are

$$\begin{aligned} \rho_0\partial_t v &= \operatorname{Div}_X P + \rho_0 b, \qquad \partial_t y = v, \qquad F = \nabla_X y, \\ \partial_t Z &= \gamma R, \\ c_0\partial_t\vartheta &= -\operatorname{Div}_X Q + r + d_v + d_Z - \beta\vartheta\operatorname{tr}(\dot{E} - \dot{Z}). \end{aligned} \tag{4.19}$$

Thus heating changes stress and internal evolution, while deformation and dissipation change temperature. Conductivity itself depends on the evolving configuration through K. All components evolve together on the same time interval.

These signs can be checked without solving a trajectory. The reference entropy is $s = -\Psi_\vartheta = \beta\,\mathrm{tr}(E-Z)+c_0\log(\vartheta/\vartheta_0)$. Consequently $\vartheta\dot{s} = -\mathrm{Div}_X\,Q+r+d_v+d_Z$, and internal energy $e = \Psi+\vartheta s$ satisfies $\dot{e} = P:\dot{F} - \mathrm{Div}_X\,Q + r$. The combined balance is therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathcal{B}}\left(\tfrac{1}{2}\rho_0|v|^2+e\right)\,\mathrm{d}X = \mathcal{P}_{\mathrm{ext}} + \int_{\mathcal{B}} r\,\mathrm{d}X - \int_{\partial\mathcal{B}} Q\cdot N\,\mathrm{d}S. \tag{4.20}$$

Viscous and internal dissipation become heat rather than disappearing from total energy. The corresponding local entropy production is $(d_v+d_Z)/\vartheta + (\nabla_X\vartheta)^T K\nabla_X\vartheta/\vartheta^2 \geq 0$ when $J,\vartheta>0$.

Several matrices within one physical mechanism. The original three-component vorticity calculation [2] already retains five blocks, B, b, s, C_2, C_1, in its two-plus-one representation. Stretching contains cross-block interactions; there is no one-matrix-per-component rule. For the solid candidate, expansion of P gives seven stress blocks:

$$\begin{aligned} P_1 &= 2\mu_s F\,\mathrm{dev}\,E, & P_2 &= \kappa_s F\,\mathrm{tr}(E)I,\\ P_3 &= -2\mu_s F\,\mathrm{dev}\,Z, & P_4 &= -\kappa_s F\,\mathrm{tr}(Z)I,\\ P_5 &= -\beta(\vartheta-\vartheta_0)F, & P_6 &= 2\eta_s F\,\mathrm{dev}\,\dot{E},\\ P_7 &= \zeta_s F\,\mathrm{tr}(\dot{E})I, & P &= \sum_{\ell=1}^{7} P_\ell. \end{aligned} \tag{4.21}$$

The blocks share F, and their state dependence is coupled through (4.19). Elasticity, internal deformation, and viscosity each occupy more than one block. Which blocks constitute the Matrix of Error is determined relative to the declared nominal model; the complete balance retains them all. For example, a nominal elastic model retains P_1+P_2, with P_3–P_7 as additional stress contributions. The number seven belongs to this chosen decomposition, not to a universal limit or a count of independent dimensions.

Integrated formulation and a concrete test. Substitute each P_ℓ separately into (4.5). Its weak contribution is $\int_I\int_{\mathcal{B}} P_\ell : \nabla_X\varphi_k\,\mathrm{d}X\,\mathrm{d}t$; stacking test windows and equation components gives the contribution arrays. Inertia, body loading, and boundary traction remain separate. For symmetric tensor tests Ξ and scalar tests χ vanishing at the temporal endpoints, the internal and heat balances give

$$-\int_I\int_{\mathcal{B}} Z:\partial_t\Xi\,\mathrm{d}X\,\mathrm{d}t - \gamma\int_I\int_{\mathcal{B}} R:\Xi\,\mathrm{d}X\,\mathrm{d}t = 0, \tag{4.22}$$

$$\begin{aligned} -c_0\int_I\int_{\mathcal{B}}\vartheta\partial_t\chi\,\mathrm{d}X\,\mathrm{d}t - \int_I\int_{\mathcal{B}} Q\cdot\nabla_X\chi\,\mathrm{d}X\,\mathrm{d}t + \int_I\int_{\Gamma_q} q_N\chi\,\mathrm{d}S\,\mathrm{d}t\\ - \int_I\int_{\mathcal{B}}\left[r+d_v+d_Z-\beta\vartheta\,\mathrm{tr}(\dot{E}-\dot{Z})\right]\chi\,\mathrm{d}X\,\mathrm{d}t = 0. \end{aligned} \tag{4.23}$$

Here $q_N = Q\cdot N$ is outward heat flux, and χ vanishes on any prescribed-temperature boundary. These equations add thermal and internal-evolution blocks beyond the seven stress blocks. Shared coefficients must remain shared when recovering parameters; fitting every block independently would define a different material. Integration removes explicit acceleration and outer divergence from these balances, while reconstruction of F, strain rate, and the internal state remains part of the observation problem.

Observability requires a separate check. The spherical elastic, internal, and thermal blocks P_2, P_4, P_5 are scalar fields times F; restricted or near-uniform heating can make their tested contributions dependent or poorly conditioned. Their tensor direction alone does not prove dependence when those scalar fields vary independently in space and time. Likewise, identifying H_s and γ requires observing or reconstructing the response of Z through its coupled evolution. The current algebra checks do not establish that these parameters are recoverable from a chosen measurement set.

Heat-driven cubic-time deformation. The thermal block supplies a calculable τ^3 contribution to deformation under a specific onset, where $\tau = t - t_0$. Take constant coefficients and reference density, no body force, and initial data $y = X$, $v = 0$, $Z = 0$, and $\vartheta = \vartheta_0$. Let $r(X, t_0) = r_0(X)$ be smooth. On a periodic reference cell, or locally in the interior with compatible boundary data, suppose a solution is sufficiently smooth for the following time and space expansions. Initially $P = 0$, $\dot{Z} = 0$, $\dot{F} = 0$, and $\partial_t^2 y = 0$. The heat and stress equations then give

$$\partial_t \vartheta(t_0) = \frac{r_0}{c_0}, \qquad \partial_t P(t_0) = -\frac{\beta r_0}{c_0} I, \qquad \partial_t^3 y(t_0) = -\frac{\beta}{\rho_0 c_0} \nabla_X r_0. \tag{4.24}$$

The viscous stress has zero initial time derivative here because both $\dot{F}$ and $\partial_t^2 F$ vanish initially. Integrating the initial acceleration growth yields

$$y(X, t_0 + \tau) - X = -\frac{\beta}{6\rho_0 c_0} \nabla_X r_0(X) \tau^3 + O(\tau^4). \tag{4.25}$$

The cubic term is in the deformation response; the heat input itself need not be cubic in time. Heat accumulation changes stress, the stress gradient produces acceleration, and its time integration produces the displacement. This is an early-time consequence of the coupled equations, not a prescribed trajectory or a claim that the complete response is a cubic polynomial.

For nonzero β, a nonzero Hessian $\nabla_X^2 r_0$ distinguishes shape change from a spatially uniform leading translation. The Green strain satisfies

$$E(X, t_0 + \tau) = -\frac{\beta}{6\rho_0 c_0} \nabla_X^2 r_0(X) \tau^3 + O(\tau^4), \tag{4.26}$$

and, with an L^2-controlled remainder on the reference body, its Lebesgue volume measure is

$$\int_{\mathcal{B}} |E|^2 \, \mathrm{d}X = \frac{\beta^2 \tau^6}{36 \rho_0^2 c_0^2} \int_{\mathcal{B}} |\nabla_X^2 r_0|^2 \, \mathrm{d}X + O(\tau^7). \tag{4.27}$$

Thus cubic-order strain and a sixth-order squared-strain integral describe the same onset. For a test φ with compact support or periodic cancellation, the initial momentum identity also reads $\rho_0 \int_{\mathcal{B}} \partial_t^3 y(t_0) \cdot \varphi \, \mathrm{d}X = (\beta/c_0) \int_{\mathcal{B}} r_0 \, \mathrm{Div}_X \, \varphi \, \mathrm{d}X$. This is the contribution of P_5 in the same integration-by-parts organization as the other stress blocks.

The calculation holds for any selected n under these assumptions. For example, $r_0 = \bar{r} + r_1 \cos(k \cdot X)$, with $r_1 \neq 0$ and $k \neq 0$, gives a leading displacement proportional to $k \sin(k \cdot X) \tau^3$ and nonzero leading strain. Choosing every component of k nonzero engages all selected coordinates; it is not required for nonzero strain. The additional choice $\bar{r} > |r_1|$ ensures positive heat supply. A local constitutive check and a linearized longitudinal-mode calculation in $n = 2, 3, 4, 7$ recover this cubic coefficient; the latter was independently checked by time integration. Uniform heating in the homogeneous periodic case or zero thermal coupling removes this contribution, while a nonzero initial mechanical acceleration can give a quadratic leading term. These checks concern the stated onset, not the full nonlinear loading response or material validation.

A full loading test. A subsequent formulation test can use a rectangular reference body in $\mathbb{R}^n$, initially at $y = X$, $v = 0$, $Z = 0$, and $\vartheta = \vartheta_0$, clamp one face, apply coupled bending and rotational traction on another, and prescribe nonuniform heat flux with a stated thermal bath on the remaining boundary. The selected n and the physical interpretation of its coordinates and loads must be specified for that application. The problem is to predict the coupled deformation during loading and after unloading and cooling. These boundary-driven conditions must be checked separately from the thermal-onset assumptions above. The single-amplitude warp remains a kinematic verification case, not the assumed solution of this test. Positivity of J and temperature, local material stability, energy closure, observability of the retained blocks, and agreement with reference calculations must be checked before claiming a predictive or computational advantage. No such full loading experiment is reported here.

4.6 The next worked deformation calculation

The next calculation should predict deformation under stated loading, using the proposed coupled system or a published material law selected for the specimen. Its parameters, temperature and strain range, initial conditions, and boundary data must be fixed before testing. Heating enters through the thermal balance and its coupling to stress and internal evolution. Thermal post-buckling experiments such as [25] provide a possible reference class, while thermoelasticity and plastic-metric models [19, 20] supply alternative constitutive settings when the proposed response is insufficient.

The full field problem should determine which components are active before a reduced description is adopted. If the one-parameter warp is used as a check, its adequacy must be tested against additional modes and material variables. Loading and unloading or cooling should be predicted, with deformation and motion measures tracked together and recovery or residual deformation compared with an independently calculable or published case. Computational cost, required observations, and prediction error should be recorded against a reference solution of the same physical model. These checks would evaluate the proposed tractability advantage; the existing warp integration does not by itself supply them.

5 Mutual Gravity for Unequal Bodies

5.1 An integral formulation for finite body count

Let $r_i(t) \in \mathbb{R}^3$ denote the position of body i, with constant mass $m_i > 0$. Define, away from collisions,

$$K_{ij} = \frac{r_j - r_i}{|r_j - r_i|^3} \quad (i \neq j), \qquad K_{ii} = 0, \qquad \ddot{r}_i = g_i = G \sum_{j \neq i} m_j K_{ij}, \qquad F_i = m_i g_i. \tag{5.1}$$

The source mass determines each contribution to acceleration; force also includes the receiving mass. All bodies attract and respond. With $D_m = \operatorname{diag}(m_1, \ldots, m_N)$, introduce the weighted graph Laplacian

$$L_{ij} = -\frac{G m_i m_j}{|r_j - r_i|^3} \ (i \neq j), \qquad L_{ii} = -\sum_{j \neq i} L_{ij}, \qquad D_m \ddot{R} = -L(R) R, \tag{5.2}$$

where the rows of R are the positions. This is a nonlinear matrix equation: the weights change with the separations. There are $N(N-1)/2$ unordered pairs, ten for five bodies.

On a collision-free interval beginning at t_0, direct time integration gives

$$v_i(t) = v_i(t_0) + G\sum_{j\neq i} m_j \int_{t_0}^{t} K_{ij}(s)\,\mathrm{d}s,$$
$$r_i(t) = r_i(t_0) + (t-t_0)v_i(t_0) + G\sum_{j\neq i} m_j \int_{t_0}^{t} (t-s)K_{ij}(s)\,\mathrm{d}s. \tag{5.3}$$

These exact coupled integral equations apply to every finite N under the stated assumptions. The unknown trajectory remains inside each kernel; the worked examples below evaluate this coupling using reference motions and numerical integration. If every separation is at least $d > 0$, then $|g_i| \le G\sum_{j\neq i} m_j/d^2$, which supplies integrability on finite intervals with that bound.

The spatial source measure is $\mu_t = \sum_j m_j \delta_{r_j(t)}$, while the trajectory integrals use Lebesgue time measure. Replacing this discrete source by a resolved density is a physical change of model, examined in Section 8.

5.2 Weak integration and correction columns

For a scalar ϕ with $\phi = \phi' = 0$ at a window's endpoints, integration by parts gives both

$$-\int v_{i\alpha}\phi'\,\mathrm{d}t = G\sum_{j\neq i} m_j \int K_{ij,\alpha}\phi\,\mathrm{d}t = \int r_{i\alpha}\phi''\,\mathrm{d}t. \tag{5.4}$$

The position-only form uses two integrations by parts. Derivatives are evaluated on the known test function, not by differencing sampled velocities or positions. With row index (w, i, α), set

$$y_{wi\alpha} = -\int v_{i\alpha}\phi'_w\,\mathrm{d}t, \qquad \Theta_{wi\alpha,j} = \int K_{ij,\alpha}\phi_w\,\mathrm{d}t, \qquad \beta_j = Gm_j, \qquad y = \Theta\beta. \tag{5.5}$$

The signed contribution array $[\,y \mid -\Theta\,\mathrm{diag}(\beta)\,]$ has row sum $y - \Theta\beta$. If a nominal description retains only some pairs, write $\Theta = \Theta_0 + \Theta_c$ and

$$\mathcal{E} = -\Theta_c\,\mathrm{diag}(\beta), \qquad r_0 = y - \Theta_0\beta, \qquad r = r_0 + \mathcal{E}\mathbf{1}. \tag{5.6}$$

The entries of $\mathcal{E}$ are the omitted gravitational contributions relative to that nominal choice. They are not integration constants.

In the main five-body example, the nominal model retains the four pairs involving body 4; that body remains free to move. The other six pairs are restored by (5.6). Both descriptions are evaluated on the same complete trajectory with the same masses. Their relative weak residuals are respectively 0.7798907 and 8.93×10^{-10}. The comparison concerns integrated balance, not percentage error in predicted positions.

5.3 Known motions and the three-body connection

Central configurations provide reference motions with accelerations proportional to displacement from the center of mass [26, 27]. For a circular two-body orbit of separation a, $\omega^2 = G(m_1+m_2)/a^3$.

For three unequal masses at an equilateral triangle of side a, subtracting their mass-weighted center gives

$$g_i = \frac{G}{a^3}\sum_{j\neq i} m_j(r_j - r_i) = -\frac{G\sum_j m_j}{a^3} r_i. \tag{5.7}$$

Rigid rotation with $\omega^2 = G\sum_j m_j/a^3$ is therefore an exact Lagrange solution. The unequal masses change the center of mass, not the identity.

The two-body reference uses $G = 1$, masses $(1.7, 0.4)$ and separation $a = 2$, starting at $(0, 0, 0)$ and $(2, 0, 0)$ before subtracting the center of mass. The three-body reference uses masses $(0.8, 1.3, 2.1)$ and vertices $(0, 0, 0)$, $(2, 0, 0)$, $(1, \sqrt{3}, 0)$ before the same centering. In both cases $v_i = \omega(-r_{i2}, r_{i1}, 0)$. All three central-configuration reference horizons in Table 2 are quarter periods, $T = \pi/(2\omega)$; the nonsymmetric horizon is separately chosen.

For the collinear family associated with Moulton, fix an ordering $x_1 < \cdots < x_N$ and $\omega > 0$. The auxiliary function

$$\Psi(x) = G\sum_{i<j}\frac{m_i m_j}{x_j - x_i} + \frac{\omega^2}{2}\sum_i m_i x_i^2 \tag{5.8}$$

diverges at collisions and at infinity. It is not the negative physical gravitational potential. Its derivatives satisfy

$$\begin{aligned} \partial_i\Psi &= m_i(g_i + \omega^2 x_i), \\ D^2\Psi(x)[h,h] &= \omega^2\sum_i m_i h_i^2 + 2G\sum_{i<j}\frac{m_i m_j (h_j - h_i)^2}{(x_j - x_i)^3} > 0 \quad (h \neq 0). \end{aligned} \tag{5.9}$$

Strict convexity gives a unique minimizer in this ordered region. There $g_i = -\omega^2 x_i$, and pair cancellation gives $\sum_i m_i x_i = 0$. Positions $(x_i, 0, 0)$ with velocities $(0, \omega x_i, 0)$ provide a rotating reference. For $G = \omega = 1$ and $m = (0.4, 1.1, 0.7, 1.8, 0.9)$, the numerically constructed coordinates are

$$x \simeq (-2.0781089793,\ -1.2209644934,\ -0.4122950648,\ 0.5287071097,\ 1.6791536471). \tag{5.10}$$

The unrounded configuration has $\max_i |g_i + x_i| = 4.44 \times 10^{-16}$. Its finite-interval use as a reference does not require stability under perturbations.

5.4 A nonsymmetric five-body calculation

The principal freely moving example uses the same five masses and all three position components. Table 1 gives inputs before subtracting the mass-weighted mean position and velocity. The interval is $0 \le t \le 0.75$, with $G = 1$.

What happens physically. Each pair pulls along its instantaneous connecting line. Those lines rotate and change length as all five objects move, so no heavy object eliminates the contributions of the others. For example, the initial acceleration of body 4 is approximately $(-0.31902, -0.12137, -0.12198)$; its force is 1.8 times that vector. The total internal force and torque cancel, although the force on each body is nonzero.

Table 1: Nonsymmetric five-body inputs. Masses are constant within a run and independently varied between runs.

Body	Mass	Initial position	Initial velocity
1	0.4	$(-2.0, 0.0, 0.1)$	$(0.10, 0.25, -0.03)$
2	1.1	$(-0.6, 1.0, 0.2)$	$(-0.30, 0.08, 0.02)$
3	0.7	$(0.8, -0.8, -0.3)$	$(0.18, -0.20, 0.03)$
4	1.8	$(2.1, 0.4, 0.6)$	$(-0.12, 0.20, -0.04)$
5	0.9	$(-0.5, -2.1, -0.5)$	$(0.20, -0.04, 0.01)$

Velocity Verlet [28] generates trajectories at 2048, 4096, and 8192 steps. The weak calculation then uses Simpson quadrature and nine translated test windows of half-width $h = 0.2T$:

$$\phi = (1 - z^2)^6, \quad \phi' = -\frac{12z(1 - z^2)^5}{h}, \quad \phi'' = \frac{12(1 - z^2)^4(11z^2 - 1)}{h^2}, \qquad z = \frac{t - t_c}{h}, \tag{5.11}$$

for $|z| < 1$, with zero continuation outside. The centers span $0.2T$ to $0.8T$. Trajectory refinement and quadrature refinement on a fixed trajectory are checked separately.

Table 2: Reference-motion checks at 8192 steps. Position errors are normalized by the initial mass-weighted RMS radius; energy drift by $K(0) + |U(0)|$.

Case	Horizon	Position error	Energy drift
Circular two-body	3.06588	1.90×10^{-8}	5.66×10^{-15}
Lagrange three-body	2.16790	1.18×10^{-8}	4.18×10^{-15}
Moulton five-body	1.57080	1.66×10^{-8}	2.99×10^{-15}
Nonsymmetric five-body	0.75000	No exact orbit assigned	1.20×10^{-10}

All four position-refinement ratios are approximately four, as expected for the second-order method in the resolved regime. The full benchmark passed 92 predeclared checks, including pair antisymmetry, zero net internal torque, relabeling invariance, known mass and length scalings, conservation diagnostics, and weak integration. The nonsymmetric example has smallest sampled separation 1.23424; sampling is a diagnostic, not a certified between-sample separation bound.

Its velocity-based and position-only weak residuals are 8.92891×10^{-10} and 8.93237×10^{-10}. With known G, the 135-by-5 mass matrix has rank five and column-scaled condition number 3.502. The maximum relative mass-recovery error is 1.34×10^{-9}. Five further runs change one mass at a time to 0.20, 1.76, 1.40, 1.26, and 1.26, respectively; their maximum recovery errors range from 4.76×10^{-9} to 8.16×10^{-9}.

These are same-model synthetic consistency checks. If G were also unknown, the equations would identify Gm_j, not independently determine G and every mass. Time-dependent mass transfer would require additional momentum terms. The finite-N identities themselves do not require equal masses, a particular ordering, or a fixed central source.

6 Multibody Dynamics: Articulated Systems

6.1 Five bodies with rotation and joint constraints

This section applies the Matrix-of-Error integration process within the established study of multibody dynamics. Liu and Jain's *A Quick Tutorial on Multibody Dynamics* [29] develops Lagrangian and Newton–Euler formulations for physics-based simulation; those mechanical formulations provide the foundation for the integrated treatment here. The example uses five massive rectangular blocks joined by ideal massless connectors and revolute joints, with the first joint fixed. The two-link Acrobot equations provide an independent published reference [30]. The calculation retains these mechanical couplings in their integrated form.

Let θ_i be each link's absolute angle measured from downward vertical. Relative joint angles are $q = R\theta$, with $q_1 = \theta_1$ and $q_i = \theta_i - \theta_{i-1}$ for $i > 1$. Use

$$m = (0.4, 1.1, 0.7, 1.8, 0.9), \qquad l = (1.00, 0.90, 1.10, 0.80, 1.05). \tag{6.1}$$

The center of block i lies at $a_i = l_i/2$ along its connector; the block's length, width, and depth are $b_i = 0.6l_i$, $w_i = 0.08l_i$, and $s_i = 0.06l_i$. For uniform density, volume integration gives

$$\rho_i = \frac{m_i}{b_i w_i s_i}, \qquad I_i = \int_{B_i} \rho_i (X^2 + Y^2)\, \mathrm{d}V = \frac{m_i(b_i^2 + w_i^2)}{12}. \tag{6.2}$$

Thus the mass and rotational inertia come from the same material measure. The model uses dimensionless external gravitational acceleration $g_e = 1$, mutual constant $G = 0.05$, damping coefficients $(0.03, 0.025, 0.02, 0.015, 0.01)$, and spring coefficients $(0.06, 0.05, 0.04, 0.03, 0.02)$. These values exercise the different contributions; they are not a terrestrial apparatus specification.

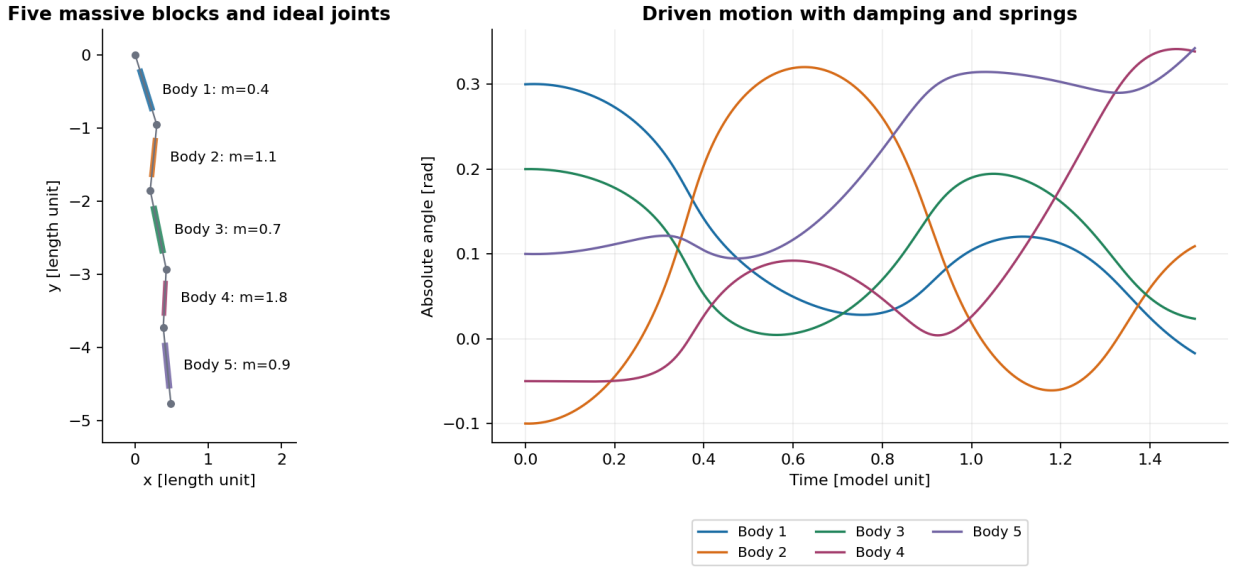


Figure 1: The supported mechanical system and an example of its coupled motion. The massive blocks occupy only part of each ideal connector. The five absolute angles evolve together under the retained inertia, gravity, damping, spring, and actuation terms.

Put $e_i = (\sin\theta_i, -\cos\theta_i)$ and

$$r_i = \sum_{k<i} l_k e_k + a_i e_i, \qquad A_{ik} = \begin{cases} l_k, & k < i, \\ a_i, & k = i, \\ 0, & k > i, \end{cases} \qquad B = A^T \operatorname{diag}(m) A. \tag{6.3}$$

The mechanical inertia matrix, denoted H to distinguish it from the contribution matrix, and its velocity term are

$$H_{jk} = B_{jk}\cos(\theta_j - \theta_k) + I_j\delta_{jk}, \qquad h_j = \sum_k B_{jk}\sin(\theta_j - \theta_k)\dot\theta_k^2. \tag{6.4}$$

With $\gamma_j = \sum_i m_i A_{ij}$, the external and center-point mutual potentials are

$$U_e = -g_e \sum_j \gamma_j \cos\theta_j, \qquad U_m = -G\sum_{i<j} \frac{m_i m_j}{|r_j - r_i|}. \tag{6.5}$$

All ten mutual pairs remain present. If $D = R^T \operatorname{diag}(d) R$, $K_s = R^T \operatorname{diag}(k) R$, and $Q = R^T\tau$ is the absolute-angle generalized torque corresponding to relative-joint actuators, then

$$H(\theta)\ddot\theta + h(\theta,\dot\theta) + \nabla U_e + \nabla U_m + D\dot\theta + K_s\theta = Q. \tag{6.6}$$

Ideal constraints are incorporated in the coordinates. The fixed support can exchange momentum with the moving bodies, but does no work at its fixed attachment point. The moving-body momentum balance therefore includes exchange with the support.

What happens physically. Moving one link displaces downstream masses. Their inertia and gravity therefore load joints upstream, while changing angles redirect the coupling between joints. Springs store mechanical energy; damping removes it; actuators supply or remove work. The error of an independent-pendulum description has an identifiable physical content: it omits these connected contributions.

6.2 Integrate generalized momentum, retaining its configuration dependence

Set $T = \frac{1}{2}\dot\theta^T H\dot\theta$ and $p = H\dot\theta$. The Euler–Lagrange equation, integrated against an endpoint-vanishing ϕ, becomes

$$-\int p_j\phi'\,\mathrm{d}t - \int \partial_{\theta_j} T\,\phi\,\mathrm{d}t + \int \left[\partial_{\theta_j}U_e + \partial_{\theta_j}U_m + (D\dot\theta)_j + (K_s\theta)_j - Q_j\right]\phi\,\mathrm{d}t = 0, \tag{6.7}$$

where

$$\partial_{\theta_j} T = -\dot\theta_j \sum_k B_{jk}\sin(\theta_j - \theta_k)\dot\theta_k. \tag{6.8}$$

Configuration dependence remains inside p and in the kinetic derivative. Pulling H outside integration as though it were constant would lose a product-rule contribution. As in the preceding examples, the operation is integration by parts with known test-function derivatives.

For the nominal reference use $p_j^0 = (m_j a_j^2 + I_j)\dot\theta_j$ and the body's own external-gravity term $m_j g_e a_j \sin\theta_j$. Table 3 spells out the eight signed columns. The five additions between the two nominal terms and actuation form the physical correction to the nominal actuated balance.

Table 3: Signed integrated columns of the articulated balance. Their sum is the numerical closure residual; the correction columns themselves retain physical contributions.

Column	Contribution for coordinate j
Nominal inertia	$-\int p_j^0 \phi' \, \mathrm{d}t$
Nominal gravity	$\int m_j g_e a_j \sin\theta_j \, \phi \, \mathrm{d}t$
Motion coupling	$-\int (p_j - p_j^0)\phi' \, \mathrm{d}t - \int \partial_{\theta_j} T \, \phi \, \mathrm{d}t$
Downstream gravity	$\int (\partial_{\theta_j} U_e - m_j g_e a_j \sin\theta_j)\phi \, \mathrm{d}t$
Mutual gravity	$\int \partial_{\theta_j} U_m \, \phi \, \mathrm{d}t$
Damping	$\int (D\dot{\theta})_j \phi \, \mathrm{d}t$
Spring	$\int (K_s \theta)_j \phi \, \mathrm{d}t$
Actuation	$-\int Q_j \phi \, \mathrm{d}t$

Thirteen test windows on $[0, 6]$, each with half-width 1.2 and the bump in (5.11), give 65 weak rows. On a prescribed smooth motion with analytically calculated required torques, Simpson quadrature gives the normalized closure $\|\mathcal{C}\mathbf{1}\|_2 \,/\, \sum_k \|\mathcal{C}_{:k}\|_2$. It falls from 2.01×10^{-13} at 1024 intervals to 3.81×10^{-16} at 4096 intervals. The nominal inertia, nominal gravity, and actuation columns alone give 0.392 on that normalization. No sampled acceleration is needed by the weak balance.

For reproducibility, with zero-based index $j = 0, \ldots, 4$, the prescribed angles are

$$\theta_{j+1}(t) = 0.08(j-2) + 0.24\sin((0.8 + 0.31j)t + 0.27j) + 0.07\cos((1.7 + 0.23j)t - 0.27j).$$

Their first two derivatives are evaluated analytically, and the required torque is given by (6.6). The window centers are $1.2 + 0.3k$, $k = 0, \ldots, 12$. They overlap and are not independent observations; local parameter rank and conditioning below describe the resulting fit, not an independent sample count.

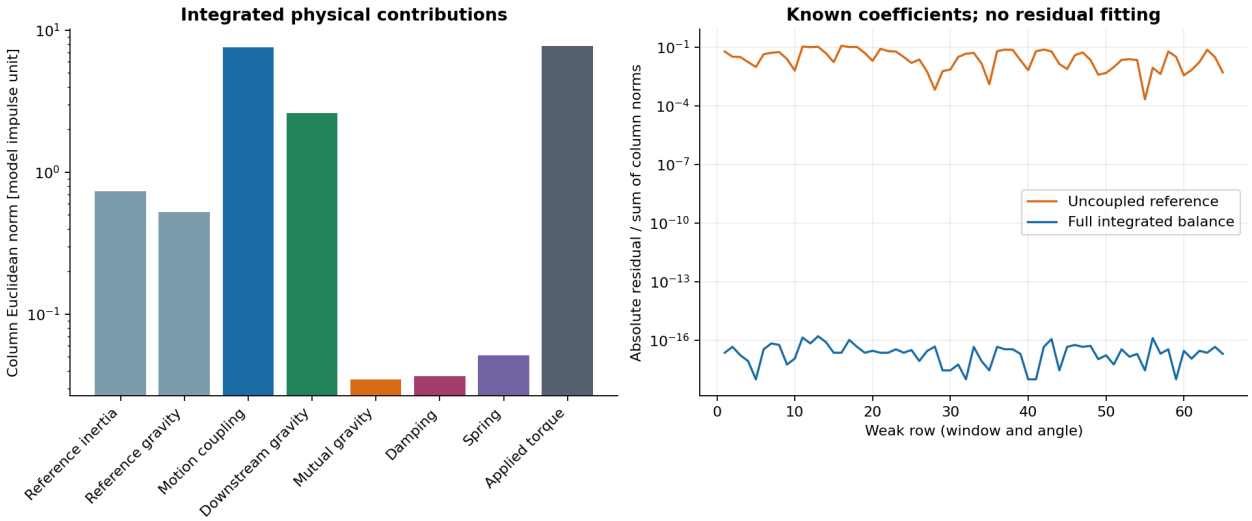


Figure 2: Integrated contributions on the same known motion. Left: column norms, which do not preserve the signs of individual rows. Right: absolute row closure normalized by the sum of column norms, with known coefficients rather than fitted cancellation. Large physical contributions can sum to a small residual.

6.3 Recovery, noise sensitivity, and a separate input

For known geometry, g_e, G, damping, springs, and calibrated actuation, the weak mass equation has the form

$$A_w m + B_w \pi(m) = b_w, \qquad \pi(m) = (m_i m_j)_{i<j}. \tag{6.9}$$

The five masses determine all ten pair products as well as every inertia. A positive-mass Gauss–Newton fit therefore uses five parameters, not ten unrelated gravitational weights. On the clean prescribed record, the local scaled Jacobian has rank five and condition number 6.84; the largest relative recovered-mass error is 4.22×10^{-15}.

Each Gauss–Newton step solves the least-squares system after normalizing the Jacobian columns by their Euclidean norms. Starting with step length one, backtracking halves it until all candidate masses are positive and the residual norm strictly decreases, with at most 24 trials per iteration and 30 iterations. A step of norm at most $10^{-12}\max(1, \|m\|_2)$ terminates the fit; a failed backtracking search retains the previous iterate. Section 8 uses the same rule from its stated three starts.

Twenty seeded trials add Gaussian noise of standard deviation 0.001 radians to angles, 0.002 per time unit to angular velocities, and 0.2% of each channel's RMS to torque. The largest relative mass error among these trials is 2.26%. This is a descriptive sensitivity test, not a confidence interval. Both the target and columns contain noise. WENDy specifically treats such errors-in-variables issues [9]. The calculations here retain the stated Gauss–Newton estimator throughout.

This observation model treats angles and angular velocities as separately measured noisy channels; it does not obtain the latter by differentiating the former. Position-only noisy observations would require a separate treatment of the configuration-dependent inertia and kinetic terms, rather than a direct reuse of the point-mass identity (5.4).

The separately specified forward tests use classical fourth-order Runge–Kutta on $[0, 1.5]$, starting at $\theta_0 = (0.3, -0.1, 0.2, -0.05, 0.1)$ and $\dot\theta_0 = (0.03, -0.02, 0.01, 0, -0.015)$. For $E = T + U_e + U_m + \frac{1}{2}\theta^T K_s \theta$, their work check is

$$E(t) - E(0) = \int_0^t \left(\dot\theta^T Q - \dot\theta^T D \dot\theta\right) \mathrm{d}s. \tag{6.10}$$

At 1024 steps the scaled maximum defect is between 8.58×10^{-11} and 1.83×10^{-10} across passive, driven, damping-only, spring-only, and combined scenarios. Five independent single-mass changes also pass the work check. A different initial state and actuator phase, evaluated with clean recovered versus original masses, gives a normalized state difference of 9.36×10^{-15} using the same solver. This tests parameter transfer within the model, not physical precision at that many digits.

6.4 Independent checks and retained refinement failures

The two-link comparison with the published Acrobot equations reconciles relative angles and pivot inertias, $I_i^{\text{pivot}} = I_i + m_i a_i^2$, before comparing terms. Inertia and velocity-term discrepancies are below 2.1×10^{-16}. A recursive Newton–Euler calculation agrees with the Lagrange torques within 6.01×10^{-16}. A finite-difference potential-gradient check, positive-definiteness checks, and a one-link small-amplitude reference supply separate tests.

The initial 256/512/1024-step mechanical refinement passed 25 of 27 checks. Passive and spring-only refinement ratios were about 27, outside the declared interval 8 to 24. A retained follow-up at 512/1024/2048/4096 steps gives finest ratios 16.05 and 16.95, consistent with fourth-order convergence. The models and thresholds were not changed. These empirical resolution checks are not interval-arithmetic error certificates.

7 A Separate Five-Body Radiation Calculation

The radiation example uses the free collinear five-body relative equilibrium in (5.10) as its isolated source, separately from the supported mechanical chain. It adds a familiar relativistic observable to a checked mechanical source under an explicit approximation.

Use physical scales $M_* = 10^{30}\,\mathrm{kg}$, $L_* = 10^9\,\mathrm{m}$, and $T_* = (L_*^3/(GM_*))^{1/2} = 3870.77\,\mathrm{s}$. The numerical constants are $G = 6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$, $c = 299792458\,\mathrm{m\,s^{-1}}$, and one kiloparsec $= 3.085677581491367 \times 10^{19}\,\mathrm{m}$; the energy-loss timescale uses a 365.25-day year. All five masses and positions are scaled from the same central configuration. Let an observer lie at distance $R_o = 1$ kiloparsec along the normal to the rotation plane. The leading mass-quadrupole approximation [31] uses the combined trace-free source

$$\mathcal{Q}_{ab} = \sum_i m_i \left(r_{ia} r_{ib} - \frac{1}{3}\delta_{ab}|r_i|^2 \right) \tag{7.1}$$

and the power and face-on strains

$$P = \frac{G}{5c^5}\sum_{ab}(\mathcal{Q}'''_{ab})^2, \qquad h_+ = \frac{G}{c^4 R_o}(\mathcal{Q}''_{xx} - \mathcal{Q}''_{yy}), \qquad h_\times = \frac{2G}{c^4 R_o}\mathcal{Q}''_{xy}. \tag{7.2}$$

Strains are evaluated at retarded source time. For the rigidly rotating collinear configuration, $S = \sum_i m_i x_i^2$ and $\omega = 1/T_*$ give the independent closed expressions

$$P = \frac{32G}{5c^5}S^2\omega^6, \qquad h_0 = \frac{4G}{c^4 R_o}S\omega^2. \tag{7.3}$$

The general calculation evaluates the derivatives from positions, velocities, accelerations, and analytic pairwise jerks. It agrees with (7.3) within 1.8×10^{-15} and also recovers the standard circular-binary expression.

What happens physically. The changing distribution of all five masses determines the signal. Power is quadratic in the combined quadrupole, so summing five independently assigned powers would discard cross terms. In this phase-locked configuration those cross terms contribute about 70.9% of the flux; the incoherent sum retains only 29.1%. This is a further example of a physical contribution that disappears if the interacting system is decomposed too aggressively.

The speeds, interbody potentials, and observer distance in Table 4 support the chosen slow-motion, weak-field, wave-zone calculation for idealized sources. The listed potential is $\max_i \sum_{j\neq i} Gm_j/(c^2|r_j - r_i|)$, not the largest single-pair value. The first post-Newtonian expansion scale v^2/c^2 is about 3.2×10^{-6}, comparable to that potential scale; it is an order estimate, not a computed correction coefficient. Radiation reaction is not inserted into the Newtonian orbit.

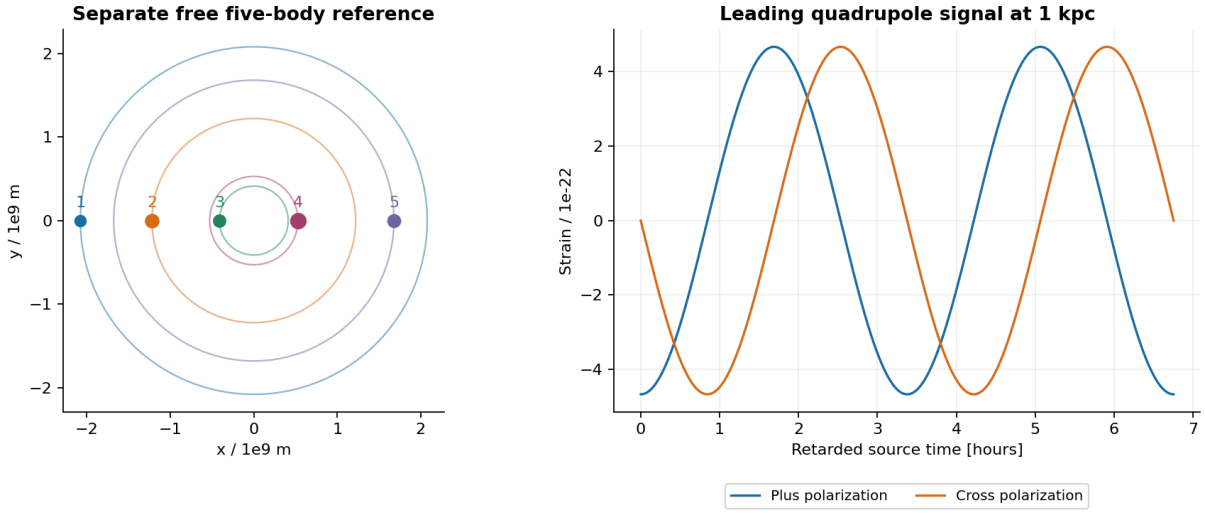


Figure 3: The separate free source and its coherent leading-quadrupole waveform. All five positions contribute to the combined source tensor before the power is calculated. The illustrative physical scale is not an observed astronomical system.

Table 4: Magnitude and approximation checks for the five-body radiation example.

Quantity	Value
Orbital period	6.76 hours
Wave frequency	8.22×10^{-5} Hz
Largest orbital speed divided by c	0.00179
Largest summed potential magnitude from the other bodies, divided by c^2	2.93×10^{-6}
Wavelength divided by observer distance	1.18×10^{-7}
Leading luminosity	2.23×10^{24} W
Face-on strain amplitude	4.67×10^{-22}
Newtonian orbital energy	-2.18×10^{41} J
Radiated energy per orbit	5.43×10^{28} J
Loss per orbit divided by orbital-energy magnitude	2.49×10^{-13}

The instantaneous ratio $|E|/P \simeq 3.09 \times 10^9$ years is therefore an energy-loss timescale, not a lifetime or merger forecast. No internal stellar structure or stability of the special configuration is assumed. The calculation establishes the leading emission and its scale while keeping the underlying approximation visible.

8 Resolved Volume, Omitted Contributions, and Inferred Mass

The preceding recovery tests use data generated from the same model being fitted. A stronger question is whether a specified missing physical contribution changes inferred parameters, and whether restoring that contribution improves results outside the fitted configurations. The following comparison answers that question within the same mechanical setting and the same estimation process.

8.1 The physical correction is a double-volume integral

For separated rigid blocks, replace the center-point mutual potential by

$$\begin{aligned} U_m^{\mathrm{vol}} &= -G \sum_{i<j} \int_{B_i} \int_{B_j} \frac{\rho_i(x)\rho_j(y)}{|y-x|} \, \mathrm{d}y \, \mathrm{d}x \\ &= -G \sum_{i<j} m_i m_j \int_{B_i} \int_{B_j} \frac{\mathrm{d}\mu_j(y) \, \mathrm{d}\mu_i(x)}{|y-x|}, \end{aligned} \tag{8.1}$$

where μ_i is the normalized uniform volume measure on block i. The generalized force is $-\nabla_\theta U_m^{\mathrm{vol}}$. Differentiation must retain both center motion and orientation; translated centers alone do not account for the torque associated with rotating the resolved mass distribution.

At the initial configuration of the mechanical example, tensor Gauss–Legendre quadrature gives a center-point potential of -0.3273237733 and an order-8 resolved potential of -0.3438376917. Relative to the resolved quantities, the potential difference is 4.8028% and the generalized-force difference is 0.51319%. The order-6-to-8 changes are 4.80×10^{-9} and 1.27×10^{-7}, respectively. These are distinguishable physical-model differences, not quadrature errors of comparable size.

The original order-4-to-6 force change, 3.14×10^{-5}, exceeded the predeclared 10^{-5} tolerance. The order-8 follow-up resolved the check without changing that tolerance. A finite-difference orientation-gradient test and a volume-inertia check provide independent confirmation of the assembly. The positive bounding-sphere clearance at this configuration is 0.353 model length units. The forward trajectories in Section 6 use point-center gravity. This audit evaluates the resolved-volume contribution at the stated configuration.

8.2 Three models and a fixed training split

The comparison retains all five bodies, all ten pairs, the same geometry, and five positive masses. It evaluates:

1. center-point gravity with the known masses;

2. center-point gravity with five fitted masses;

3. resolved-volume gravity with five fitted masses.

Both fitted models constrain pair products through the same five masses. Reference data use spatial quadrature order 10; the resolved candidate uses order 8. No arbitrary pair weights are added to improve the incomplete model.

The configuration experiment uses 12 training states, giving 60 generalized-gravity components, and eight held-out states, giving 40 components. Training angles are sampled in $[-0.45, 0.45]$ and held-out angles in $[-0.75, 0.75]$ using the fixed seed 20261026. All configurations are retained; the smallest bounding-sphere clearance is about 0.218. The observations are gradients of the mutual potential, whose negatives are generalized forces. Norms below stack all components before normalization.

Specifically, NumPy 1.24.4's `Generator(PCG64(20261026))` draws a 12×5 uniform training array followed by an 8×5 uniform wider-range array, in row-major order. These distributions and draw order fix the inputs. This original held-out test combines transfer with a wider angle range; Section 8.7 adds a separate test within the training range.

The positive-mass fit uses three initial vectors:

$$(1, 1, 1, 1, 1), \qquad (0.2, 2, 0.4, 1.2, 0.6), \qquad (1.4, 0.3, 1.8, 0.5, 1.1).$$

Each fit permits at most 30 Gauss–Newton iterations and 24 backtracking steps. Selection uses training residual only. The starts converge consistently under the recorded tolerance. This noise-free synthetic comparison holds the fitting process fixed to isolate the effect of the physical model.

Table 5: Configuration-level gravity comparison. Errors are relative fractions, not percentages; the mass column is the largest relative error over the five masses.

Model	Training	Held out	Mass error
Point centers, known masses	5.827×10^{-3}	1.2149×10^{-2}	0 (fixed)
Point centers, fitted masses	3.768×10^{-3}	1.2239×10^{-2}	3.2052×10^{-2}
Resolved volume, fitted masses	6.38×10^{-11}	5.80×10^{-10}	5.55×10^{-10}

8.3 Omitted structure becomes a biased mass estimate

The fitted point masses are approximately

$$\widehat{m}_{\mathrm{pt}} = (0.394574, 1.102458, 0.678114, 1.857693, 0.878479). \tag{8.2}$$

Relative to the known masses, their changes are -1.357%, $+0.223\%$, -3.127%, $+3.205\%$, and -2.391%. Training improves, but the held-out gravity error does not. The bias has both signs, so omitted finite structure does not produce a universal apparent excess of mass. Potential data were not fitted: their held-out discrepancy is 4.59% for known point masses, 4.76% for fitted point masses, and 3.92×10^{-10} for the resolved fit.

Even within training, the point fit leaves relative error 3.768×10^{-3} instead of the known-mass 5.827×10^{-3}. Thus changing masses absorbs only part of the shape contribution; the model mismatch is already present before extrapolation is tested.

The resolved result is close to the observed numerical resolution: the order-8/order-10 held-out gravity change at the true masses is 2.65×10^{-10}. It is not an experimental precision claim. Both candidate sensitivities have rank five; their column-scaled condition numbers are approximately 20.6 and 22.1. These checks support local identifiability in this experiment, not global uniqueness for all mass-inference problems.

8.4 Carry the comparison into the full weak mechanics

A second stage uses different prescribed training and held-out angular histories, with analytic velocities and accelerations and the required resolved-model torque. Each history supplies three test windows, centered at 0.6, 1.5, and 2.4, with half-width 0.6, giving 15 rows. Time quadrature orders 12 and 20 are compared independently of the spatial quadrature. This stage evaluates the weak mechanical balance; it does not solve a new resolved-body forward trajectory.

On $[0, 3]$, training uses the prescribed motion in Section 6. For zero-based $j = 0, \dots, 4$, the held-out motion is

$$\theta_{j+1}(t) = 0.08(j-2) + o_j + 0.20\sin((0.9 + 0.27j)t + 0.4 + 0.33j) + 0.06\cos((1.4 + 0.29j)t - 0.2j),$$

where $o = (0.08, -0.06, 0.04, -0.08, 0.06)$. Differentiate these expressions analytically. Use Gauss–Legendre nodes mapped to each window, with the bump (5.11); order 20 supplies Tables 6–7. At each node the reference torque is (6.6) with order-10 resolved gravity. Uniform block quadrature uses the product of one-dimensional Gauss–Legendre weights divided by eight for each normalized body measure, with all node pairs retained. This specifies both the driving data and the integrated fitting problem without private trajectory files.

The fitted equation remains (6.9). The matrix A_w contains inertia, kinetic coupling, and external gravity. Damping and springs are known, and the sole candidate-model change is point versus volume mutual gravity in B_w. The same mass must therefore account for several physically different observations at once.

Table 6: Full weak mechanical comparison. Training and held-out errors normalize the stacked balance residual by the corresponding target norm.

Model	Training	Held out	Mass error
Point centers, known masses	1.07×10^{-5}	1.74×10^{-5}	0 (fixed)
Point centers, fitted masses	6.85×10^{-6}	2.00×10^{-5}	1.97×10^{-3}
Resolved volume, fitted masses	5.15×10^{-13}	1.32×10^{-12}	7.82×10^{-11}

The point fit now has maximum mass bias about 0.197%, smaller than the gravity-only 3.205%, because inertia and external gravity constrain the same parameters. Nevertheless, its held-out balance is about 14.8% worse than that of the point model with known masses. A small whole-balance error can conceal the omitted contribution when mutual gravity is relatively weak. Normalizing by the held-out mutual-gravity norm instead gives Table 7.

The point fit slightly improves its gravity component while worsening the total held-out balance. The resolved fit is consistent with the observed refinement levels. The held-out order-12/order-20 time change is 3.98×10^{-11}; the gravity change from volume order 8 to 10 is 5.41×10^{-10}. Reference

Table 7: The same held-out mechanical comparison on the scale of mutual gravity. Changing a mass changes the entire balance, not only its gravitational component.

Model	Gravity-component error	Balance / gravity norm
Point centers, known masses	0.007276	0.007276
Point centers, fitted masses	0.007179	0.008353
Resolved volume, fitted masses	5.45×10^{-10}	5.53×10^{-10}

weak closure is about 4.40×10^{-16}. These are empirical refinement and closure diagnostics, not independently certified physical error bounds. Both fitted local Jacobians have rank five and condition number about 18.1.

8.5 An exact account of the parameter shift

The distinction can be written without a small-perturbation approximation. Let

$$f_{\mathrm{pt}}(m) = Am + B_{\mathrm{pt}}\pi(m), \qquad d = b - f_{\mathrm{pt}}(m), \qquad \Delta m = \widehat{m} - m, \tag{8.3}$$

where m denotes the true mass vector, b the reference observations, and $J_{\mathrm{pt}}(m)$ the Jacobian of f_{pt}. Then the fitted residual $r = f_{\mathrm{pt}}(\widehat{m}) - b$ satisfies

$$r = J_{\mathrm{pt}}(m)\Delta m + B_{\mathrm{pt}}\pi(\Delta m) - d. \tag{8.4}$$

This identity is exact because each pair term expands as $(m_i + \Delta m_i)(m_j + \Delta m_j) = m_i m_j + m_i \Delta m_j + m_j \Delta m_i + \Delta m_i \Delta m_j$. With the resolved reference,

$$d = (B_{\mathrm{ref}} - B_{\mathrm{pt}})\pi(m) + \epsilon_{\mathrm{quad}}, \qquad \epsilon_{\mathrm{quad}} = b - Am - B_{\mathrm{ref}}\pi(m). \tag{8.5}$$

The volume-minus-point term is the identified physical contribution. The remaining numerical closure is zero by construction in the static comparison and near roundoff in the reference weak balance. A fit can absorb part of the former into Δm, but the held-out test reveals whether that substitution remains adequate in different configurations.

What happens physically. Two bodies with the same total mass and center do not exert exactly the same force if their extended shapes and orientations differ. A center-point model can compensate by assigning altered masses in the configurations it has seen. Those altered masses then change inertia and other forces as well, and need not work elsewhere. The integrated correction keeps the physical distribution explicit instead of asking a parameter to represent several different effects.

8.6 Verification record

Verification includes independent aggregate cubature, potential-gradient differences, rotations, the shrinking-body point limit, non-overlap, local rank, training-only selection, quadrature refinement, and exact reconstruction of (8.4). The private calculation records retain the individual checks and outcomes. Two original window jobs stopped at the unchanged 180-second runtime guard. Their failure records were retained, and the same integrals were recomputed in smaller batches without changing inputs or acceptance tolerances. No stopped job is counted as a successful result.

8.7 Second-moment correction and an independent in-range test

Two additional checks assess whether a lower-order physical correction is sufficient and whether mass refitting transfers within the training range. They supplement the retained three-model comparison without changing its inputs or results. For each uniform block, let

$$\Sigma_i = R_i \,\mathrm{diag}(b_i^2, w_i^2, s_i^2) R_i^T/12, \qquad r_{ij} = r_j - r_i, \quad d_{ij} = |r_{ij}|,$$

where R_i has columns $(e_i, e_i^\perp, e_z)$, with the planar vectors embedded in $\mathbb{R}^3$ and $e_i^\perp = (\cos\theta_i, \sin\theta_i, 0)$. The center-of-mass second-moment approximation is

$$U_m^{(2)} = -G \sum_{i<j} m_i m_j \left[\frac{1}{d_{ij}} + \frac{3 r_{ij}^T (\Sigma_i + \Sigma_j) r_{ij} - d_{ij}^2 \,\mathrm{tr}(\Sigma_i + \Sigma_j)}{2 d_{ij}^5} \right]. \tag{8.6}$$

This is the familiar monopole-plus-quadrupole contribution expressed through known body moments [36]. It retains the leading monopole–quadrupole interactions, not the higher-order quadrupole–quadrupole term. Both centers and orientations are differentiated when forming the generalized forces. No correction amplitude is fitted. This static gravitational quadrupole is distinct from the radiating source tensor in Section 7.

At the original configuration, $U_m^{(2)} = -0.3415312566$, leaving a 0.67079% potential discrepancy relative to resolved volume, compared with 4.8028% for point centers. An inertia-tensor evaluation independently reproduces the potential. Centered differences check the analytic gradient; common rotation, pair interchange, and shrinking-body controls also pass the predeclared tolerances. Potential accuracy alone is not used to infer mass accuracy.

The extra holdout consists of eight configurations drawn independently using NumPy's PCG64 generator with seed 20261027, each of the 40 entries uniform on $[-0.45, 0.45]$ in row-major order. All are retained, with positive bounding-sphere clearances. Reference and resolved-candidate quadrature remain orders 10 and 8. The second-moment model is fitted on the original twelve training configurations from the same three positive starts, using the same solver and training-only selection. Existing fitted masses are reused unchanged. No model is refitted on either holdout.

Table 8: Additional configuration comparison. Errors are relative stacked gravity-gradient norms. The in-range holdout is independent of training; the wider holdout is the original one. Mass error is the largest relative error over five masses.

Model	In range	Wider range	Mass error
Point centers, known masses	7.4945×10^{-3}	1.2149×10^{-2}	0 (fixed)
Point centers, fitted masses	5.9940×10^{-3}	1.2239×10^{-2}	3.2052×10^{-2}
Second moments, known masses	3.5892×10^{-4}	3.3449×10^{-4}	0 (fixed)
Second moments, fitted masses	1.4628×10^{-4}	1.5082×10^{-4}	2.3562×10^{-3}
Resolved volume, fitted masses	1.1751×10^{-10}	5.8007×10^{-10}	5.5452×10^{-10}

The second-moment fit gives

$$\widehat{m}_{(2)} \simeq (0.400097, 1.099815, 0.698351, 1.803223, 0.899142).$$

Its training gravity error is 1.0757×10^{-4}, compared with 4.6047×10^{-4} at the known masses. The maximum mass error falls from the point fit's 3.205% to 0.236%, but does not vanish. Its unfitted

potential errors are 0.67744% in range and 0.65977% on the wider holdout. The new in-range order-8/order-10 gravity change is 3.10×10^{-10}, well below the truncated-model discrepancies.

The new holdout qualifies the original transfer result: point-mass refitting improves gravity predictions within the sampled angle range, while failing to improve the wider-range test. The known second-moment correction improves both, and its fitted form improves them further. The persistent discrepancy and parameter bias identify the model inadequacy, while their size and transfer depend on the observations and retained physical contributions. The follow-up concerns configurations only; it adds neither a resolved forward trajectory nor a new noise experiment.

9 Connections, Reproducibility, and Further Applications

9.1 What the calculations add to the series

The methods bridge [1] remains the general guide. The present sequel adds worked applications: a weighted budget with an equality case; a volume measure of warping; a finite-N gravitational balance; a constrained mechanical balance with configuration-dependent inertia; and a controlled demonstration of parameter bias caused by omitted body structure. Table 9 distinguishes their outputs.

Table 9: Completed outputs and their immediate interpretation.

System	Completed calculation	Meaning of the result
Periodic fluid budget	Equality case, quantitative loss, threshold time	Constraint on the stated budget and its valid interval
Warping configuration	Exact strain-volume integral and rotation-distance bounds	Shape distortion at prescribed amplitude; predictive material response remains to be tested
Point-mass gravity	Finite-N integral identities and checked unequal-mass examples	All pair contributions retained and identifiable masses recovered in the tested records
Articulated mechanics	Momentum weak form, eight signed columns, work and reference checks	Inertia, gravity, constraints, and inputs organized in one integrated balance
Five-body radiation	Combined leading-quadrupole signal and flux	Coherent source contribution under the stated weak-field approximation
Resolved-volume comparison	Point, second-moment, and resolved-volume comparisons	Distribution can bias masses; correction and transfer depend on the retained model and tested range

Across these examples, the number of coordinates or bodies changes the integrands and the contribution list, not the integration-by-parts rule. Increasing model complexity still requires the appropriate equations, material assumptions, and observations. Further applications can follow the same process by specifying the interactions and evaluating their integrated contributions. The articulated example treats rigid bodies connected by joints and springs; solid deformation is developed separately in Section 4.

9.2 Density, mass discrepancy, and the mass-gap direction

Gravitational Density [32] previously carried a specified density and stress, with conservation and initial data, through a gravitational geometry to a light-travel timing integral. *Extended Relativity* [33] explained the interpretation motivating that work and the associated Matrix-of-Error framework: physical relationships are considered within one interconnected spacetime. The present calculations continue that explanatory aim using established mechanical laws and explicit approximations. They do not require a new physical law to organize the interactions.

The resolved-volume comparison supplies a controlled connection to the question of *mass discrepancy*: how much of an inferred mass difference can instead be attributed to a missing physical contribution or an insufficient description of its distribution? The answer here is specific to five known blocks. The bias has both signs, and no astronomical data enter the experiment. A later application would need the same independently specified distribution and dynamics to account for more than one observable, such as motion and propagation, rather than defining a correction from the discrepancy it is meant to explain. Agreement of effective behavior alone does not select an intrinsic origin of density.

For well-separated bodies, the leading nonspherical correction has relative potential scale of order $(R/d)^2$, multiplied by shape-dependent moments; R is a body-size scale and d the separation. Its size must be evaluated for the actual distribution and geometry, not transferred from these nearby blocks to galaxies. Extended-body astronomy already studies the dependence of such approximations on shape and separation [37]. The connection made here is a controlled method for testing an inferred-parameter discrepancy, not evidence that finite-body corrections explain an astronomical mass discrepancy.

The immediate obstacle in the *Yang–Mills mass-gap* direction is that a positive Poincare constant on a fixed bounded domain need not remain bounded away from zero as the domain expands. Equality rigidity alone therefore does not supply a uniform positive gap. The Yang–Mills question concerns a positive spectral separation above the vacuum in the appropriate quantum field theory [34], rather than a mass discrepancy inferred from gravitational observations.

A concrete successor problem is to seek a model-specific budget inequality whose equality characterization persists under domain enlargement and whose decay constant has a strictly positive lower bound independent of domain size, in fixed physical units. Connecting such an estimate to Yang–Mills would also require establishing the relevant quantum theory and the relation between the controlled quantity and its energy spectrum; a classical decay estimate alone does not supply that connection. This identifies the missing steps toward the mass-gap question and a direction for subsequent physics-focused work. The programme may yield useful intermediate results without reaching a Yang–Mills existence or mass-gap proof.

9.3 Reproducibility and numerical scope

The companion directories listed below are the author's *private working records*, not publicly accessible supplementary files. They preserve protocols, inputs, code hashes, outputs, and failed attempts:

- `part3_worked_example`: finite-body inputs, reference motions, weak matrices, and diagnostic regressions;
- `part4_worked_example`: articulated mechanics, refinement, resolved-volume audit, and separate radiation;
- `physical_model_comparison`: fixed split, three physical models, quadrature records, mass fits, and independent verification; its separately retained follow-up records contain the second-moment and in-range checks.
- `solid_proposal_investigation`: thermal-onset analysis, linearized mode, and controls. Local constitutive checks use `verify_solid_proposal.py` in the manuscript folder.

The text specifies every model input and numerical prescription needed to reproduce the tables, including reference configurations, prescribed motions, physical constants, test functions, quadrature orders, random generators and draw order, fitting starts, and stopping rules. Access to the private directories is not required to reconstruct those tabulated experiments. Independent floating-point implementations may differ in roundoff-level diagnostics; the quoted numerical precision is not a promise of bitwise-identical output. No public software or data deposit accompanies this manuscript.

One diagnostic correction is important for reuse. An all-at-rest input makes a normalization based only on initial velocities vanish. The revised point-body runner records raw conservation drifts and uses characteristic scales based on kinetic plus absolute potential energy, with explicit absolute and relative tolerances. Its cold-start regression gives raw momentum drift about 1.4×10^{-16} and angular-momentum drift 5.55×10^{-16}, rather than reporting those roundoff-sized values as enormous relative failures. The principal reference results remain preserved. Interrupted-run metadata are also checkpointed so a resource stop cannot silently erase a failed or incomplete attempt.

Each numerical test evaluates a specific part of the calculation. Known-orbit comparison checks trajectories. Weak closure checks integrated balances. Local rank checks parameter sensitivity for selected observations. A prescribed-motion torque test is not a forward-trajectory prediction. Held-out physical-model comparisons test transfer beyond the fitted configurations. None of these synthetic checks, alone, is laboratory or astronomical validation. Their combination provides a reproducible route to such an application with the omitted physical terms kept visible.

10 Conclusion

The common structure is the integration process: specify the measures, interactions, and boundary assumptions, then retain and compare the resulting contributions while distinguishing model discrepancy from numerical error. The budget yields an equality-rigidity consequence and a conditional finite threshold time. The deformation example supplies an exact nonlinear volume calculation. The multibody examples extend the same integration process from pair gravity to constrained mechanics. The model comparison demonstrates how an omitted contribution can be absorbed into fitted parameters: point-mass refitting improves the independent in-range gravity test but not the wider-range one, while the known second-moment correction reduces both prediction error and mass bias. The fully resolved model remains consistent with the observed quadrature resolution.

Building on the earlier connection between quantum measurement records, probability measures, and Lebesgue integration, this study applies that shared mathematical foundation to envelope constraints, deformation measures, and interacting-body mechanics. The worked examples extend the Matrix-of-Error framework by retaining identifiable physical contributions and demonstrating how their omission can alter inferred parameters. Together, these calculations carry the quantum-to-calculus connection into further mathematical applications, preparing the foundation for a subsequent physics-focused study.

Preparation and Provenance

The Matrix-of-Error interpretation and the research direction originate with the author. The author selected the investigation of Lebesgue measure, rigidity, solid deformation, and unequal-mass multibody systems as a continuation of the preceding work, and directed the distinction between the envelope example and the separate material-deformation question. The author also selected the physical-model comparison and the continuing quantum-to-calculus connection as part of the series' purpose.

AI assistance supported literature retrieval, algebraic and numerical implementation, source and result checking, and manuscript assembly. Separate AI-assisted review prompted conservation-diagnostic and interrupted-record corrections and the controlled omitted-physics experiment. A subsequent Fable 5.1 review reported independent checks of the original closed-form results and selected numerical values, and prompted targeted attribution and reporting clarifications. The additional second-moment and in-range configuration tests were performed afterward. A later Fable 5.1 review reported independent reproduction of those tests and of the results in Tables 6 and 7. Established theorems, equations, and methods remain attributed to their published sources. The manuscript was assembled from the retained research notes and calculation outputs; no unpublished conference lecture or poster material was supplied or used as evidence.

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